

Pathways into Complexity

Lineages, practices, and methods for working in complex systems, with cases, organizations, and live resources from the Systems Change Learning Guide.

How to read this. Each pathway opens with a one-line view of the space and its **kind** (lineage, practice, method, structure, field, tradition, or hybrid). **Road** is the lineage in rough order. **Mix in** adds adjacent voices. **Policy lens** shows how it applies to public policy and government innovation. **Engage** links to living communities, **Go deeper** names organization profiles, and **Cases** point to real situations. The companion app adds live resources from the Systems Change Learning Guide.

Orientations

Our orientations, each drawn from a lineage on this map. Use them as questions at hard junctions, not as badges, a way of holding orientations we learned from the Systems Change Learning Guide by Jaya Ramchandani and Raisa Mirza.

Consent before command. Move when no one has a reasoned objection, rather than waiting for everyone to agree or for someone to order it.

Ask: Who has not yet been asked whether they can live with this? From Sociocracy (Kees Boeke, Gerard Endenburg).

Match the variety. A response has to be as varied as the situation it meets. Do not simplify people to fit the system.

Ask: Where are we forcing many different lives through one channel? From W. Ross Ashby, requisite variety.

Widen the choices. Act so that the people affected have more options afterward, not fewer.

Ask: Does this close doors for anyone who was not in the room? From Heinz von Foerster, the ethical imperative.

Take only what is given. Relationship before extraction. Ask, take what is offered, share it, and give something back.

Ask: What are we taking from this community, and what are we returning? From Robin Wall Kimmerer, the Honorable Harvest.

Many contexts, no master frame. Every issue lives in family, work, body, place, and history at once. No single discipline or worldview holds all of it.

Ask: Which context are we ignoring because it does not fit our frame? From Nora Bateson (transcontextuality), Arturo Escobar (the pluriverse).

Learning over proving. Treat results as information for the next move, and hold people accountable for learning, not for hitting numbers they do not control.

Ask: What did we learn that changes what we do next? From Human Learning Systems, developmental evaluation, Deming.

Power in plain sight. Name who decides, whose issues never reach the table, and which beliefs make the arrangement feel natural.

Ask: Which form of power is doing the work here without being named? From Steven Lukes, John Gaventa, Paulo Freire.

Kinds of path

Lineage	A body of ideas that changes how you understand systems. It informs action without prescribing it.
Practice	A way of being together that shifts perception and relationship. It works indirectly and breaks when aimed at an output.
Method	A structured process with steps and artifacts, aimed at a designed outcome. It can be commissioned and reported on.
Structure	An arrangement of authority, decision-making, and feedback. It changes who decides and how information flows.
Field	A domain where lineages, practices, methods, and structures meet a context. It is where you work, not how.
Tradition	A whole way of knowing held by peoples and places. It enters through relationship, consent, and shared power.
Hybrid	A pathway that is several kinds at once. Watch for the method part being taken and the rest dropped.

Foundations and science

Cybernetics, the shared foundation Lineage

Steering through feedback: how systems regulate themselves, and how observers are part of what they observe.

Road: Norbert Wiener → W. Ross Ashby (requisite variety) → Heinz von Foerster (second-order cybernetics) → Maturana and Varela → Gordon Pask (conversation theory) → Paul Pangaro

Mix in: Gregory Bateson and Margaret Mead at the Macy Conferences, Stafford Beer; Yağmur Denizhan

Policy lens: Requisite variety as a design test: services must match the variety of people's lives. Beer's Project Cybersyn is the classic, contested case of cybernetic government.

Engage: [American Society for Cybernetics](#) · [The Cybernetics Society](#)

Go deeper: [American Society for Cybernetics](#)

Cases: Project Cybersyn (see Cases)

The formal science of complex systems Lineage

Mathematical and computational study of emergence, networks, adaptation, and self-organization across physics, biology, and society.

Road: Ilya Prigogine (dissipative structures) → Santa Fe Institute → network science (Barabási, Strogatz, Watts) → NECSI (Yaneer Bar-Yam)

Mix in: Hermann Haken (†2024, synergetics), Stuart Kauffman; Melanie Mitchell, Geoffrey West, Carlos Gershenson, Pierpaolo Andriani (exaptation), Jean Boulton (process complexity), Paul Cilliers, Michael Garfield

Policy lens: Agent-based models and network analysis of service systems, epidemics, and markets to test policy before rollout.

Engage: [Santa Fe Institute](#) · [Complexity Explorer \(free courses\)](#) · [Network Science textbook \(free\)](#) · [NECSI](#)

Go deeper: [Santa Fe Institute](#)

Cases: Artificial Anasazi (see Cases)

Systems thinking traditions: hard, soft, and critical Lineage

The family of approaches most people mean by systems thinking. Hard systems engineer a known goal; soft systems help people with different views learn their way into action; critical systems ask whose interests a system serves and who is left out.

Road: C. West Churchman (the systems approach) → Russell Ackoff (messes and interactive planning) → Peter Checkland (soft systems methodology) → Werner Ulrich (critical systems heuristics) → Michael C. Jackson (critical systems thinking)

Mix in: Stafford Beer, Donella Meadows, W. Edwards Deming, John Seddon (systems thinking in public services)

Policy lens: Rich pictures and stakeholder learning before policy design, asking who should benefit and who decides, and redesigning public services around demand rather than targets.

Engage: [Soft systems methodology \(overview\)](#) · [Critical systems thinking \(overview\)](#) · [Michael C. Jackson: Critical Systems Thinking and the Management of Complexity \(Wiley\)](#)

Go deeper: [The Complexity Lounge](#)

Cases: Systems thinking in Portsmouth housing repairs (see Cases)

Mind, life, and constraints Lineage

How living things and minds hold themselves together and act with purpose. Constraints, not forces, shape what can happen next; organisms keep making the conditions of their own existence; brains act to keep their predictions about the world true.

Road: Robert Rosen (life as a system closed to efficient causation) → Francisco Varela and Evan Thompson (enaction, *The Embodied Mind*) → Alicia Juarrero (*Dynamics in Action, Context Changes Everything*: constraints as causes) → Andy Clark (the predictive mind) → Karl Friston (the free energy principle and active inference) → Matteo Mossio (*Biological Autonomy*, with Alvaro Moreno)

Mix in: Humberto Maturana, Terrence Deacon, Ezequiel Di Paolo (*Linguistic Bodies*), biosemiotics, cybernetics

Policy lens: Mostly a source of concepts rather than a policy tool: designing by changing constraints rather than issuing instructions, and treating people as active, sense-making agents rather than targets of messages.

Engage: [Free energy principle \(overview\)](#) · [Enactivism \(overview\)](#) · [Moreno and Mossio: Biological Autonomy \(Springer\)](#)

Go deeper: [The Complexity Lounge](#)

Cases: An EU field guide for managing crises with constraints (see Cases)

Relational, Indigenous, and meaning

Relational, transcontextual, and post-humanist knowing Lineage

Knowing through relationships rather than parts. Attends to how contexts shape each other and questions the human-centered, modern frame itself.

Road: Gregory Bateson (Steps to an Ecology of Mind) → Nora Bateson (Warm Data, Combining) → Bayo Akomolafe (These Wilds Beyond Our Fences) → Vanessa Machado de Oliveira (Hospicing Modernity)

Mix in: Tyson Yunkaporta (Sand Talk), Donna Haraway (Staying with the Trouble), Karen Barad (agential realism), Isabelle Stengers. Joanna Macy (†2025) bridged systems theory, Buddhism, and deep ecology; Anna Tsing, Rosi Braidotti, María Puig de la Bellacasa, Bruno Latour

Policy lens: Shows how policy categories fragment lived lives. Warm Data Labs with public teams, and the practice of hospicing institutions that no longer serve.

Engage: [International Bateson Institute](#) · [Warm Data Lab](#) · [The Emergence Network](#) · [Bayo Akomolafe](#) · [Gesturing Towards Decolonial Futures](#) · [Work That Reconnects Network](#)

Go deeper: [International Bateson Institute](#) · [The Emergence Network](#) · [Gesturing Towards Decolonial Futures](#)

Cases: Te Awa Tupua, the Whanganui River (see Cases)

Nora Bateson and Warm Data Hybrid

Writer, filmmaker, and President of the International Bateson Institute. She carries her father Gregory's ecology of mind into living practice through Warm Data: information about the interrelationships that bind a complex system, found only by moving across contexts.

Road: Gregory Bateson → An Ecology of Mind (film, 2010) → Small Arcs of Larger Circles (2016) → Warm Data Labs and host trainings → Combining (2023). Key ideas: transcontextual inquiry, symmathesy (mutual learning within living systems), aphanipoiesis (unseen coalescence toward vitality).

Mix in: Bayo Akomolafe, Tyson Yunkaporta, Otto Scharmer (as a structured counterpoint), the hosting practices

Policy lens: Warm Data Labs hosted with public staff and residents change how people inside the system perceive and relate to an issue. Whatever policy or design follows comes from that shifted ground in its own time, not from findings harvested and handed over.

Engage: [International Bateson Institute](#) · [Warm Data Lab](#)

Go deeper: [International Bateson Institute](#)

Cases: Zero Step Warm Data Project (see Cases)

Indigenous and land-based knowledge Tradition

Knowledge held in relationship with land, kin, and story, where reciprocity and pattern are themselves ways of knowing.

Road: Vine Deloria Jr. → Gregory Cajete (Native Science) → Robin Wall Kimmerer (Braiding Sweetgrass, the Honorable Harvest) → Tyson Yunkaporta (Sand Talk)

Mix in: Leroy Little Bear, Winona LaDuke, Elinor Ostrom on the commons

Policy lens: Indigenous-led stewardship and co-management of land and water, reciprocity as a design principle, and consultation that shares power instead of extracting input.

Engage: [Robin Wall Kimmerer](#) · [Center for Native Peoples and the Environment](#)

Go deeper: [Center for Native Peoples and the Environment](#)

Cases: Menominee Forest · Te Awa Tupua, the Whanganui River (see Cases)

Presencing, Theory U, and Social Presencing Theater Hybrid

Change that starts by suspending habitual seeing, sensing the whole, and letting new possibilities take form through the group.

Road: Kurt Lewin (action research) → Argyris and Schön → Peter Senge → Otto Scharmer (Theory U, shaped in part by his interviews with Francisco Varela) → Arawana Hayashi (Social Presencing Theater, sensing a system through embodied movement)

Mix in: David Bohm (On Dialogue), Katrin Kaufer, Adam Kahane (transformative scenario planning)

Policy lens: Co-sensing with residents and frontline staff before anything is designed. u.lab runs are widely used by public teams. Social Presencing Theater is hosted as embodied sensing, not as a technique for gathering input.

Engage: [Presencing Institute and u.lab](#) · [Reos Partners](#)

Go deeper: [Presencing Institute](#) · [Reos Partners](#)

Cases: Sustainable Food Lab (see Cases)

The metacrisis and meaning Lineage

Treats interlocking ecological, social, and epistemic crises as one pattern rooted in how modern culture perceives and makes meaning.

Road: Bateson and Edgar Morin (†2026) → Iain McGilchrist (The Matter with Things) → Jonathan Rowson (Perspectiva) → Zak Stein and John Vervaeke

Mix in: Vanessa Machado de Oliveira, Nora Bateson

Policy lens: Long-horizon governance, and the sensemaking and meaning-making capacity democracies need to handle intertwined crises.

Engage: [Perspectiva](#) · [Channel McGilchrist](#)

Go deeper: [Gesturing Towards Decolonial Futures](#) · [Perspectiva](#)

Cases: Well-being of Future Generations Act (see Cases)

Semiotics: signs, codes, and meaning-making Lineage

How anything comes to mean something: words, images, rules, forms, tracks in the snow.

Meaning is not stored in things but made in a three-way relation between a sign, what it points to, and whoever reads it, and whole cultures are woven from these relations.

Road: Charles Sanders Peirce (icon, index, symbol; semiosis) → Ferdinand de Saussure (signifier and signified) → Yuri Lotman (the semiosphere, Tartu-Moscow school) → Umberto Eco (*A Theory of Semiotics*) → Eduardo Kohn (*How Forests Think*, semiotics beyond the human)

Mix in: Roland Barthes (myth), Thomas Sebeok, Gregory Bateson, the narrative change pathway, biosemiotics

Policy lens: Public signs, forms, and warnings are sign systems. Semiotics helps check what a message actually signals to different readers, design warnings meant to outlast the people who made them, and see how official categories shape what a system can notice.

Engage: [Peirce's theory of signs \(Stanford Encyclopedia of Philosophy\)](#) · [Semiosphere \(overview\)](#) · [Eduardo Kohn: How Forests Think \(UC Press\)](#) · [University of Tartu: Institute of Philosophy and Semiotics](#)

Go deeper: [University of Tartu Department of Semiotics](#) · [International Association for Semiotic Studies](#)

Cases: Warning people 10,000 years from now (see Cases)

Living systems

Living systems and regeneration Lineage

Life as self-producing and interdependent, applied to ecological resilience and regenerative design of cultures and economies.

Road: Maturana and Varela (autopoiesis) → Fritjof Capra → Daniel Christian Wahl (Designing Regenerative Cultures), alongside C.S. Holling → Carl Folke (resilience)

Mix in: Donella Meadows, Joanna Macy; Bill Mollison and David Holmgren (permaculture), Daniel Walsh, Zita Cobb, Christina Bowen

Policy lens: Resilience planning, climate adaptation, and bioregional thinking that treats places as living systems.

Engage: [Capra Course](#) · [Stockholm Resilience Centre](#)

Go deeper: [Stockholm Resilience Centre](#) · [Capra Course](#)

Cases: Menominee Forest · European mink on Hiiumaa and Saaremaa (see Cases)

Biosemiotics and the Umwelt Lineage

Life runs on signs, not just chemistry. Every organism lives in its own Umwelt, the world as its senses and needs make it meaningful, and living systems are webs of interpretation from cells to ecosystems.

Road: Jakob von Uexküll (*Umwelt, A Foray into the Worlds of Animals and Humans*) → Thomas Sebeok (zoosemiotics) → Jesper Hoffmeyer (*Signs of Meaning in the Universe*, semiotic freedom) → Kalevi Kull (Tartu biosemiotics, ecosemiotics) → Terrence Deacon (*Incomplete Nature*)

Mix in: Gregory Bateson (a difference that makes a difference), Maturana and Varela, Charles Sanders Peirce, Eduardo Kohn, Timo Maran (ecosemiotics)

Policy lens: Conservation and urban planning that ask what a place means to the species living there, not only how many there are: Umwelt-based analysis of city green spaces, and reintroduction projects that read local human meanings of an animal as carefully as its ecology.

Engage: [International Society for Biosemiotic Studies](#) · [Biosemiotics \(overview\)](#) · [Umwelt \(overview\)](#) · [Ecosemiotics at the University of Tartu](#)

Go deeper: [International Society for Biosemiotic Studies](#) · [University of Tartu Department of Semiotics](#)

Cases: European mink on Hiiumaa and Saaremaa (see Cases)

Organizing and sensemaking

Sensemaking and acting in uncertainty Method

Practical frameworks for telling ordered problems from complex ones and acting through small, safe-to-fail probes.

Road: W. Ross Ashby → Karl Weick (†2026) → Alicia Juarrero → Dave Snowden (Cynefin) → Glenda Eoyang (Human Systems Dynamics)

Mix in: Stacey and Shaw's complex responsive processes; Sonja Blignaut, Tony Quinlan, Zhen Goh, Michael Cheveldave, Mary Boone, Viv Read and Julie Cunningham, John Boyd (the OODA loop), Trent Hone, Riva Greenberg

Policy lens: Cynefin in policy and national risk work, safe-to-fail probes instead of big pilots, and distributed citizen narratives as live evidence.

Engage: [The Cynefin Company](#) · [Human Systems Dynamics Institute](#)

Go deeper: [The Cynefin Company](#) · [Human Systems Dynamics Institute](#) · [The Complexity Lounge](#)

Cases: Risk Assessment and Horizon Scanning · An EU field guide for managing crises with constraints (see Cases)

Governance and self-organizing structures Structure

Designing decision structures that use feedback, consent, and distributed authority instead of command and control.

Road: Ashby → Stafford Beer (Viable System Model) → Kees Boeke and Gerard Endenburg (†2025, sociocracy) → Elinor Ostrom (polycentric governance)

Mix in: Frederic Laloux, the Buurtzorg case, Indy Johar (Dark Matter Labs)

Policy lens: Ostrom's design principles for co-governance of shared resources, polycentric arrangements across agencies, and consent-based structures inside teams.

Engage: [Sociocracy For All](#) · [Ostrom Workshop](#) · [American Society for Cybernetics](#)

Go deeper: [Sociocracy For All](#) · [Ostrom Workshop](#)

Cases: Project Cybersyn · Buurtzorg neighbourhood nursing (see Cases)

Complex responsive processes Lineage

Organizations are not systems that managers stand outside and design. They are patterns of conversation and power between people, continually reproduced and changed in everyday interaction. Change happens in local relating, not in plans.

Road: Norbert Elias (figurations: people bound together in shifting webs of interdependence) → George Herbert Mead (the self arises in social gesture and response) → Ralph Stacey (complex responsive processes of relating, with Patricia Shaw and Douglas Griffin) → Patricia Shaw (changing conversations in organizations) → Christopher Mowles (reflexive practice, Doctor of Management at Hertfordshire) → Chris Rodgers (informal coalitions)

Mix in: Karl Weick, Group Relations, Bohm Dialogue, Cynefin

Policy lens: Treating policy and reform as conversations that shift what people take for granted, paying attention to power and anxiety in meetings, and building reflection into how public managers learn.

Engage: [Complexity and Management blog \(University of Hertfordshire staff group\)](#) · [Ralph D. Stacey \(overview\)](#) · [Christopher Mowles, University of Hertfordshire](#)

Go deeper: [The Complexity Lounge](#) · [Complexity and Management \(University of Hertfordshire\)](#)

Cases: Chronic illness care in an Australian general practice (see Cases)

Movements and hosting

Emergence in movements and social change Field

How small, relational, adaptive actions scale into systemic change without central control.

Road: Grace Lee Boggs and Octavia Butler → adrienne maree brown (Emergent Strategy) → Margaret Wheatley and the Berkana Two Loops model

Mix in: Nora Bateson on change that happens between contexts

Policy lens: Governments hosting rather than directing: funding networks, backing community-led pilots, and using the Two Loops model to steward transitions.

Engage: [adrienne maree brown](#) · [Berkana Institute](#) · [Margaret Wheatley](#)

Go deeper: [Berkana Institute](#) · [James and Grace Lee Boggs Center](#)

Cases: Student Volunteer Army · Highlander Folk School and the civil rights movement · Wajir Peace and Development Committee (see Cases)

Hosting and participatory practice Practice

Methods for convening groups so that collective intelligence and new patterns can emerge from the room.

Road: Harrison Owen (Open Space) → Juanita Brown and David Isaacs (World Café) → Art of Hosting (Toke Møller, Chris Corrigan) → Liberating Structures (Lipmanowicz and McCandless) → Warm Data Labs

Mix in: David Bohm (On Dialogue), sociocracy, Social Presencing Theater; Chris Corrigan

Policy lens: Participatory processes, citizens' assemblies, and deliberative forums, with Art of Hosting methods long used in public institutions.

Engage: [Art of Hosting](#) · [Liberating Structures](#) · [The World Café](#)

Go deeper: [Art of Hosting](#) · [Liberating Structures](#)

Cases: Participatory budgeting (see Cases)

Living practices

Bohm Dialogue Practice

Groups suspending assumptions and attending to collective thought as it moves, with no agenda or decision to reach.

Road: David Bohm (On Dialogue) → William Isaacs (Dialogue and the Art of Thinking Together) → Peter Senge's learning organization work

Mix in: Otto Scharmer, Nora Bateson, the hosting practices

Policy lens: Dialogue held for its own sake in cross-agency and community settings, with no decision on the table. Over time it loosens the assumptions that would otherwise harden into policy.

Engage: practiced widely, with no single home

Read and watch: [Bohm Dialogue \(overview\)](#) · [The art of dialogue \(MIT, on William Isaacs's Dialogue Project\)](#) · [David Bohm: On Dialogue \(Routledge\)](#)

Cases: Dialogue at a steel mill (see Cases)

T-groups and sensitivity training Practice

Small unstructured groups learning from their own here-and-now interaction. The root of much of organization development.

Road: Kurt Lewin → NTL Institute at Bethel, Maine (1947) → Chris Argyris and Edgar Schein (†2023) → Stanford's Interpersonal Dynamics course

Mix in: Group Relations, action research

Policy lens: Leadership development that builds capacity to notice group dynamics, power, and feedback inside public teams.

Engage: [NTL Institute](#)

Go deeper: [NTL Institute](#)

Cases: The New Britain workshop (see Cases)

Group Relations Practice

Temporary learning institutions where members study authority, role, and unconscious group dynamics as they happen.

Road: Wilfred Bion (Experiences in Groups) → A.K. Rice and the Tavistock Institute's Leicester Conference (since 1957) → the A.K. Rice Institute in the US

Mix in: T-groups, systems psychodynamics

Policy lens: Understanding authority and role in bureaucracies, and why systems unconsciously resist the change they commission.

Engage: [Tavistock Institute](#)

Go deeper: [Tavistock Institute of Human Relations](#)

Cases: The Leicester Conference (see Cases)

Appreciative Inquiry Method

Change that starts from what already gives life to a system rather than from its problems.

Road: David Cooperrider and Suresh Srivastva (Case Western, 1980s) → the 4-D cycle → Appreciative Inquiry Summits for whole-system change

Mix in: Positive deviance, World Café

Policy lens: Whole-system summits with residents, staff, and partners that design from existing strengths.

Engage: practiced widely, with no single home

Go deeper: [Fowler Center, Case Western Reserve University](#)

Cases: Imagine Chicago (see Cases)

Work That Reconnects Practice

A group practice that moves through gratitude, honoring pain for the world, seeing with new eyes, and going forth.

Road: Joanna Macy (†2025) → Molly Young Brown and facilitators worldwide → the Work That Reconnects Network

Mix in: Deep ecology, the relational pathway

Policy lens: Sustaining public servants and communities working on climate and loss, where despair blocks action.

Engage: [Work That Reconnects Network](#)

Go deeper: [Work That Reconnects Network](#)

Cases: Despair and empowerment work after Chernobyl (see Cases)

Somatic and embodied practice Practice

Change that lives in the body: how bodies hold history, shape what we can perceive under stress, and settle enough to act together. Treats embodiment as a condition for working in complexity, not a wellness add-on.

Road: Wilhelm Reich and body-oriented psychology → Richard Strozzi-Heckler (somatic leadership) → Staci K. Haines (generative somatics, *The Politics of Trauma*) → Resmaa Menakem (*My Grandmother's Hands*, cultural somatics)

Mix in: Arawana Hayashi (Social Presencing Theater), Joanna Macy, the crisis and transition cluster

Policy lens: Trauma-informed services and workplaces, frontline staff regulation in high-stress roles, and noticing how bodies in a room shape what a public meeting can actually hear.

Engage: [Staci K. Haines: What is social justice somatics?](#) · [generative somatics legacy archive](#) · [Resmaa Menakem](#)

Go deeper: [Cultural Somatics Institute and generative somatics](#)

Cases: The Trojan Warrior Project (see Cases)

Power, narrative, and conflict

Power analysis Method

Making power visible: who decides, whose issues never reach the table, and which beliefs make an arrangement feel natural. The missing question in much systems work.

Road: Steven Lukes (*Power: A Radical View*, three faces of power) → John Gaventa (Power Cube: spaces, places, and forms of power) → Lisa VeneKlasen and Valerie Miller (*A New Weave of Power*, JASS)

Mix in: Paulo Freire, Michel Foucault, Sasha Costanza-Chock, Group Relations

Policy lens: Mapping who actually shapes a policy, which consultations are closed, invited, or claimed spaces, and where invisible power (norms, beliefs) keeps residents out.

Engage: [Powercube \(IDS\)](#) · [JASS: Just Power guide](#)

Go deeper: [Institute of Development Studies: Powercube](#) · [JASS \(Just Associates\)](#)

Cases: Participatory budgeting (see Cases)

Narrative change Method

Shifting the deep stories a society uses to make sense of an issue, so different policies become thinkable. Works over years through many voices, not one campaign.

Road: George Lakoff (framing) → FrameWorks Institute (strategic frame analysis) → Narrative Initiative (*Toward New Gravity*) → Marshall Ganz (public narrative: story of self, us, and now)

Mix in: adrienne maree brown, Bayo Akomolafe, Hilary Cottam

Policy lens: Reframing public problems so structural causes become visible, and giving public servants and residents a way to tell a shared story of why change is needed now.

Engage: [FrameWorks Institute: Five questions about framing](#) · [Narrative Initiative: Toward New Gravity](#) · [Leading Change Network: Public Narrative](#)

Go deeper: [FrameWorks Institute](#) · [Narrative Initiative](#) · [Leading Change Network](#)

Cases: Reframing Aging (see Cases)

Conflict transformation Hybrid

Treating conflict as information about a system and a doorway into changed relationships, rather than a problem to settle quickly. Holds justice and relationship together.

Road: John Paul Lederach (*The Moral Imagination*) → Arnold and Amy Mindell (Deep Democracy, Worldwork) → Howard Zehr (restorative justice) → Marshall Rosenberg (Nonviolent Communication)

Mix in: Group Relations, T-groups, the somatic pathway, adrienne maree brown

Policy lens: Restorative approaches in justice, schools, and housing; community mediation; and hosting charged public meetings so minority voices are not steamrolled by consensus.

Engage: [Kroc Institute: The Moral Imagination](#) · [Process Work Institute: Worldwork](#) · [Zehr Institute for Restorative Justice](#)

Go deeper: [Kroc Institute for International Peace Studies](#) · [Process Work Institute](#) · [Zehr Institute for Restorative Justice](#)

Cases: Wajir Peace and Development Committee · Truth and Reconciliation Commission · Dialogue at a steel mill (see Cases)

Technology, AI, and complexity Field

What happens when automated systems meet complex social life. Models trained on the past freeze patterns in place, measure what is easy rather than what matters, and fall hardest on people who are already marginalized, so the question is who is harmed, not only how accurate the model is.

Road: Norbert Wiener (warnings about automation) → Abeba Birhane (relational ethics, complexity and machine learning) → Timnit Gebru and Emily M. Bender (the stochastic parrots paper) → Kate Crawford (*Atlas of AI*)

Mix in: John Willis on Deming and AI, Dave Snowden and Simon Wardley on AI, power analysis, measurement and accountability

Policy lens: Impact assessment and redress before automated decisions touch benefits, policing or health; banning protected characteristics as risk factors; keeping humans accountable for decisions a model informs.

Engage: [Birhane and Cummins: Algorithmic Injustices, towards a relational ethics \(arXiv\)](#) · [Distributed AI Research Institute \(DAIR\)](#) · [Amnesty International: Xenophobic machines \(2021\)](#)

Go deeper: [Distributed AI Research Institute \(DAIR\)](#)

Cases: The Dutch childcare benefits scandal (see Cases)

Crisis and transition

Crisis, disaster, and high reliability Field

What people and organizations do when normal order breaks. Disasters tend to produce mutual aid, not panic, and the best-run high-risk organizations stay alert to small failures instead of trusting their plans.

Road: Enrico Quarantelli (disaster sociology) → Karl Weick and Kathleen Sutcliffe (*Managing the Unexpected*, high reliability organizing) → Rebecca Solnit (*A Paradise Built in Hell*) → Daniel Aldrich (social capital in recovery) → Dean Spade (mutual aid)

Mix in: Cynefin's chaotic domain, the somatic pathway, Ostrom on self-organized commons

Policy lens: Emergency management that works with spontaneous volunteers rather than against them, investing in neighborhood ties before disasters, and after-action learning without blame.

Engage: [Natural Hazards Center](#) · [Weick and Sutcliffe: high reliability principles](#) · [Dean Spade on mutual aid \(interview\)](#)

Go deeper: [Natural Hazards Center](#)

Cases: Student Volunteer Army · The loss of the space shuttle Columbia (see Cases)

Ritual, liminality, and transition Lineage

Why ritual matters when systems change. Transitions pass through a threshold where old roles dissolve before new ones form, and communities have long held people through that in-between.

Road: Arnold van Gennep (rites of passage) → Victor Turner (liminality, communitas) → William Bridges (endings, neutral zone, new beginnings) → Francis Weller (grief as communal work)

Mix in: Work That Reconnects, Bayo Akomolafe, Vanessa Machado de Oliveira (hospicing), Indigenous and land-based knowledge

Policy lens: Marking endings when programs close or teams merge, grief and memorial work after a crisis, and treating reorganizations as transitions people move through, not org charts.

Engage: [William Bridges Associates: What is transition?](#) · [Francis Weller: The Geography of Sorrow \(The Sun\)](#)

Go deeper: [The Emergence Network](#) · [Gesturing Towards Decolonial Futures](#) · [Work That Reconnects Network](#) · [William Bridges Associates](#)

Cases: Truth and Reconciliation Commission (see Cases)

Resilience engineering and safety science Field

Why things usually go right, not only why they go wrong. Safety comes from people adapting to messy, changing work, so organizations learn more from everyday work and near misses than from blame after accidents.

Road: Diane Vaughan (normalization of deviance) → Erik Hollnagel (Safety-II: learning from what goes right) → David Woods (resilience engineering) → Richard Cook (*How Complex Systems Fail*) → Sidney Dekker (just culture, drift into failure) → Ron Westrum (a typology of organizational cultures)

Mix in: Karl Weick and high reliability organizing, Gary Klein (naturalistic decision making), Cynefin's work on safety, the crisis pathway

Policy lens: Incident reviews that ask how work is really done, not who to blame; regulators and health systems moving from counting errors to learning from everyday adaptation.

Engage: [Richard Cook: How Complex Systems Fail](#) · [Hollnagel, Wears and Braithwaite: From Safety-I to Safety-II, a white paper \(NHS England, PDF\)](#) · [DORA: generative organizational culture \(Westrum's typology\)](#)

Go deeper: [The Complexity Lounge](#) · [DORA \(DevOps Research and Assessment\)](#)

Cases: The loss of the space shuttle Columbia (see Cases)

Policy, systemic design, and economics

Public policy and government innovation Field

Moving public management from targets and linear delivery toward learning, relationships, experimentation, and adaptive stewardship.

Road: Rittel and Webber (wicked problems) → Donella Meadows (leverage points) → Geoff Mulgan (collective intelligence, public imagination) → Toby Lowe (Human Learning Systems) → Hilary Cottam (Radical Help, the relational state)

Mix in: Mariana Mazzucato (mission-oriented policy), Christian Bason (design for policy), Indy Johar (Dark Matter Labs), Bill Sharpe (Three Horizons), UNDP's portfolio approach to systems change; Thea Snow (relational government)

Policy lens: This is the home field for the policy lens. Read it alongside the Strategy, money, and accountability cluster, where budgets, contracts, and metrics decide whether any of this survives contact with a real institution.

Engage: [Human Learning Systems](#) · [Centre for Public Impact](#) · [UCL Institute for Innovation and Public Purpose](#) · [Nesta](#) · [UNDP Accelerator Labs](#) · [International Futures Forum](#) · [H3Uni \(Three Horizons\)](#) · [OECD Observatory of Public Sector Innovation](#) · [Dark Matter Labs](#)

Go deeper: [Human Learning Systems](#) · [Centre for Public Impact](#) · [UCL Institute for Innovation and Public Purpose](#) · [Nesta](#) · [UNDP Accelerator Labs](#) · [Dark Matter Labs](#)

Cases: Amsterdam City Doughnut · Plymouth Alliance for Complex Needs · Human Learning Systems in practice · Legible Practises · People's Plan Campaign · Well-being of Future Generations Act (see Cases)

Systemic design and futures Method

Design practice scaled up to whole systems, paired with ways of using the future to act differently in the present.

Road: Horst Rittel → Birger Sevaldson (gigamapping) → Hugh Dubberly and Paul Pangaro → Riel Miller (futures literacy) → Bill Sharpe (Three Horizons)

Mix in: Dan Hill (Dark Matter and Trojan Horses), Christian Bason, Dunne and Raby (critical design), Stuart Candy (experiential futures)

Policy lens: Government policy labs, anticipatory governance, futures literacy for civil servants, and portfolio approaches instead of single-bet programs.

Engage: [Systemic Design Association](#) · [UNESCO Futures Literacy](#) · [H3Uni \(Three Horizons\)](#)

Go deeper: [International Futures Forum](#) · [UNESCO Futures Literacy](#)

Cases: Mont Fleur scenarios (see Cases)

Complexity economics and new economic models Lineage

The economy as an evolving, adaptive system of people and institutions rather than an equilibrium machine.

Road: Herbert Simon (bounded rationality) → W. Brian Arthur (increasing returns) → Eric Beinhocker (The Origin of Wealth) → Kate Raworth (Doughnut Economics)

Mix in: J. Doynne Farmer (Making Sense of Chaos), Elinor Ostrom, Mariana Mazzucato

Policy lens: City Doughnut portraits (Amsterdam adopted the model in 2020), mission-oriented innovation, and regulation that expects adaptation.

Engage: [Doughnut Economics Action Lab](#) · [INET Oxford](#)

Go deeper: [Ostrom Workshop](#) · [UCL Institute for Innovation and Public Purpose](#) · [Doughnut Economics Action Lab](#)

Cases: Amsterdam City Doughnut (see Cases)

Strategy, money, and accountability

Strategy under uncertainty Method

Strategy that assumes the future cannot be predicted: rehearse several futures, map how value chains evolve, and build options that benefit from surprise.

Road: Pierre Wack (Shell scenarios) → Adam Kahane (transformative scenario planning) → Simon Wardley (Wardley Mapping) → Nassim Nicholas Taleb (antifragility)

Mix in: Three Horizons, Cynefin, futures literacy

Policy lens: Scenario work before long-lived infrastructure or budget commitments, and mapping which public capabilities are commodity versus genuinely new before procuring them.

Engage: [Pierre Wack: Scenarios, Uncharted Waters Ahead \(HBR\)](#) · [Learn Wardley Mapping: Simon's book](#) · [Adam Kahane: Transformative Scenario Planning \(SSIR excerpt\)](#)

Go deeper: [International Futures Forum](#) · [Reos Partners](#) · [Wardley Mapping community](#)

Cases: Risk Assessment and Horizon Scanning · Mont Fleur scenarios · Warning people 10,000 years from now (see Cases)

Funding and budgeting for adaptation Structure

Money that can follow learning. Replaces fixed annual budgets and single big bets with rolling allocation, portfolios of experiments, and funders who trust grantees.

Road: Jeremy Hope and Robin Fraser (Beyond Budgeting) → Bjarte Bogsnes (Equinor) → UNDP portfolio approach → Trust-Based Philanthropy Project

Mix in: Human Learning Systems, developmental evaluation, Mariana Mazzucato

Policy lens: Multi-year flexible funding, portfolio budgets for missions, and grant terms that let providers change course when they learn something.

Engage: [Bogsnes: Beyond Budgeting at Equinor \(Drucker Forum\)](#) · [Trust-Based Philanthropy practices](#) · [UNDP: portfolio approach to system transformation](#)

Go deeper: [UNDP Accelerator Labs](#) · [Trust-Based Philanthropy Project](#)

Cases: Plymouth Alliance for Complex Needs · Handelsbanken and Equinor without annual budgets (see Cases)

Measurement and accountability Lineage

Why targets distort what they measure, and what accountability can look like when outcomes are emergent. Moves from proving results to learning in public.

Road: Donald Campbell and Charles Goodhart (the laws named after them) → Mark Moore (public value, the strategic triangle) → Jerry Muller (*The Tyranny of Metrics*) → Toby Lowe (accountability for learning)

Mix in: Deming, developmental evaluation, Human Learning Systems

Policy lens: Replacing target regimes with learning reviews, reporting what changed and why, and holding services accountable for the health of relationships and systems, not activity counts.

Engage: [Jerry Muller: The Tyranny of Metrics \(Chronicle\)](#) · [Moore's strategic triangle \(Atlas of Public Management\)](#)

Go deeper: [Human Learning Systems](#) · [The W. Edwards Deming Institute](#)

Cases: Human Learning Systems in practice · Strive Partnership · The Dutch childcare benefits scandal (see Cases)

Government plumbing Field

The rules, contracts, and routines where reluctant bureaucracies actually push back. Procurement, delivery, regulation, and frontline discretion decide whether complexity methods survive.

Road: Michael Lipsky (*Street-Level Bureaucracy*) → Charles Sabel and Jonathan Zeitlin (experimentalist governance) → relational commissioning (Plymouth Alliance) → Jennifer Pahlka (*Recoding America*)

Mix in: Nesta on anticipatory regulation and sandboxes, Hilary Cottam, service design

Policy lens: Alliance contracts instead of payment by results, policy written with delivery teams at the table, regulatory sandboxes, and trusting frontline discretion instead of scripting it away.

Engage: [Recoding America](#) · [Sabel and Zeitlin: Experimentalist Governance \(chapter\)](#) · [Nesta: Renewing Regulation](#)

Go deeper: [Human Learning Systems](#) · [Nesta](#) · [Recoding America](#)

Cases: Plymouth Alliance for Complex Needs · Government Digital Service and GOV.UK · Systems thinking in Portsmouth housing repairs (see Cases)

Teams and learning cultures Method

The team-level conditions that let people surface bad news, learn fast, and organize around flow instead of hierarchy.

Road: Amy Edmondson (psychological safety, teaming) → Jeff Gothelf and Josh Seiden (*Sense and Respond*) → Matthew Skelton and Manuel Pais (*Team Topologies*)

Mix in: Deming, sociocracy, T-groups; Nelson Repenning (Dynamic Work Design), Mary Uhl-Bien (complexity leadership), Gemma Jiang and Barry Johnson (polarity thinking), Tiani Jones

Policy lens: Blameless reviews in public agencies, product teams that own a service end to end, and outcomes over outputs in digital delivery.

Engage: [Amy Edmondson: Building a psychologically safe workplace \(TEDx\)](#) · [Team Topologies key concepts](#)

Go deeper: [Team Topologies](#)

Cases: Buurtzorg neighbourhood nursing · A decade of research on software delivery (see Cases)

Corporate innovation and renewal Field

How established organizations explore new value while running the current business, and when they need to break apart into smaller self-managing units.

Road: Clayton Christensen (disruptive innovation) → Charles O'Reilly and Michael Tushman (ambidextrous organization) → Henry Chesbrough (open innovation) → Haier (RenDanHeYi microenterprises)

Mix in: Buurtzorg, Beyond Budgeting, Wardley Mapping

Policy lens: Public agencies face the same exploit and explore tension: protected space for new services without starving the ones residents rely on today.

Engage: [What Is Disruptive Innovation? \(HBR\)](#) · [The Ambidextrous Organization \(HBR\)](#) · [Corporate Rebels on RenDanHeYi](#)

Go deeper: [Corporate Rebels](#)

Cases: Buurtzorg neighbourhood nursing · Handelsbanken and Equinor without annual budgets · Haier's RenDanHeYi (see Cases)

Collaboration across boundaries Method

How many organizations work on one shared problem without one of them owning it. The field has moved from shared measurement toward equity and community power.

Road: John Kania and Mark Kramer (collective impact, 2011) → Mark Cabaj and Liz Weaver (Collective Impact 3.0) → Collective Impact Forum's equity turn

Mix in: Ostrom, network weaving, Two Loops, Human Learning Systems

Policy lens: Cross-agency place-based partnerships, shared learning agendas instead of shared KPIs, and residents holding real seats in governance.

Engage: [Collective Impact 3.0 \(Collective Impact Forum\)](#)

Go deeper: [Tamarack Institute](#) · [Collective Impact Forum](#)

Cases: Sustainable Food Lab · Strive Partnership (see Cases)

Design and practice

Design thinking Method

Framing problems through empathy, prototyping, and iteration. Strong for bounded problems, weaker where no single solution exists.

Road: Herbert Simon (The Sciences of the Artificial) → Horst Rittel → Nigel Cross (Designing Ways of Knowing) → Donald Schön (reflection-in-action) → IDEO and the Stanford d.school

Mix in: Natasha Iskander's critique of design thinking

Policy lens: Human-centered problem framing in agency innovation teams. Most useful when paired with complexity methods for problems without one right answer.

Engage: [Stanford d.school](#) · [IDEO](#)

Go deeper: [Stanford d.school](#)

Cases: Embrace infant warmer (see Cases)

Service design Method

Services as journeys across touchpoints, with frontstage experience and backstage systems designed together.

Road: Lynn Shostack (service blueprinting) → Birgit Mager (Köln) → Marc Stickdorn (This Is Service Design Doing) → Lou Downe (Good Services)

Mix in: UK Government Digital Service, US Digital Service, 18F (closed 2025); Jen Briselli and Kyle Godbey (complexity in service design)

Policy lens: End-to-end benefit and permit journeys, service standards, and blueprints that expose the policy and system constraints behind a bad experience.

Engage: [Service Design Network](#) · [Good Services](#)

Go deeper: [Good Services](#)

Cases: Government Digital Service and GOV.UK (see Cases)

Participatory design and design justice Method

Designing with people rather than for them, with power named and shared in the process.

Road: Scandinavian participatory design (Pelle Ehn, Kristen Nygaard) → Elizabeth Sanders (co-design) → Ezio Manzini (Design, When Everybody Designs) → Sasha Costanza-Chock (Design Justice) → Arturo Escobar (Designs for the Pluriverse)

Mix in: The relational and Indigenous pathways; Christopher Alexander (pattern languages), Jen Briselli and Kyle Godbey, Dori Tunstall (decolonizing design)

Policy lens: Co-designing programs with the residents they serve, with explicit attention to who holds decision power.

Engage: [Design Justice Network](#) · [DESIS Network](#)

Go deeper: [Design Justice Network](#)

Cases: Participatory budgeting (see Cases)

Transition design Method

Design for long-horizon societal transitions, grounded in living systems theory. The design path most explicitly built on complexity.

Road: Terry Irwin, Gideon Kossoff, and Cameron Tonkinwise (Carnegie Mellon)

Mix in: Three Horizons, Capra, Meadows

Policy lens: Long-horizon transitions in energy, food, and care that span agencies and election cycles.

Engage: [Transition Design](#)

Go deeper: [Transition Design \(Carnegie Mellon\)](#)

Cases: Water shortage in Ojai (see Cases)

Strategic design and policy labs Field

Using design to reshape institutions and the rules, budgets, and culture underneath them.

Road: Helsinki Design Lab (Sitra, Bryan Boyer, Dan Hill) and MindLab (Christian Bason) → Policy Lab UK and La 27e Région → Dark Matter Labs

Mix in: Geoff Mulgan, Nesta

Policy lens: In-house policy labs and stewardship models that change procurement, budgeting, and governance, not just services.

Engage: [Policy Lab UK](#) · [La 27e Région](#) · [Dark Matter Labs](#)

Go deeper: [Dark Matter Labs](#) · [Helsinki Design Lab \(Sitra, archived\)](#)

Cases: Legible Practises (see Cases)

Improvement and adaptive management Method

Learning cycles built into daily operations, treating the organization as a system to be continually improved.

Road: W. Edwards Deming (System of Profound Knowledge) → Toyota and Lean → Don Berwick and IHI (PDSA cycles) → Agile

Mix in: Plexus Institute's healthcare work, Brenda Zimmerman; Trisha Greenhalgh and Paul Plsek (complexity in health care), John Willis

Policy lens: PDSA cycles in public health and service delivery, and continuous improvement units inside agencies.

Engage: [The Deming Institute](#) · [Institute for Healthcare Improvement](#)

Go deeper: [The W. Edwards Deming Institute](#) · [Institute for Healthcare Improvement](#)

Cases: 100,000 Lives Campaign (see Cases)

Developmental evaluation Method

Evaluation built for innovation in complex settings, feeding real-time learning instead of judging afterward.

Road: Michael Quinn Patton → Mark Cabaj and the Tamarack Institute

Mix in: Human Learning Systems, Cynefin

Policy lens: Funding and accountability for adaptive programs, where evaluators help programs learn while they run.

Engage: [Tamarack Institute](#) · [BetterEvaluation](#)

Go deeper: [Human Learning Systems](#) · [Tamarack Institute](#)

Cases: Human Learning Systems in practice (see Cases)

Action research Method

Inquiry and change as one act, led with and by the people affected.

Road: Kurt Lewin → Paulo Freire and Orlando Fals-Borda (participatory action research) → Peter Reason and Hilary Bradbury

Mix in: Chris Argyris (action science), the hosting pathway

Policy lens: Community-led inquiry into local problems, with residents as co-researchers and co-owners of findings.

Engage: [Action Research Network of the Americas](#)

Go deeper: [Action Research Network of the Americas](#)

Cases: Participatory action research on Colombia's Caribbean coast (see Cases)

Positive deviance Method

Finding the uncommon practices that already work inside a community and helping them spread.

Road: Jerry and Monique Sternin

Mix in: Plexus Institute, which championed it in healthcare

Policy lens: Scaling what already works in the field, from frontline caseworkers to communities, before importing outside solutions.

Engage: [Positive Deviance Collaborative](#)

Go deeper: [Positive Deviance Collaborative](#)

Cases: Child nutrition through positive deviance (see Cases)

Software, flow, and architecture Hybrid

Complexity as it shows up in building software: work flows or gets stuck, forecasts beat estimates, culture predicts delivery, and architectures should be designed to survive stresses nobody predicted.

Road: Taiichi Ohno (the Toyota Production System) → W. Edwards Deming (systems of profound knowledge) → Donald Reinertsen (*The Principles of Product Development Flow*) → Nicole Forsgren, Jez Humble and Gene Kim (*Accelerate*, DORA research) → Barry O'Reilly (residuality theory)

Mix in: Simon Wardley, Troy Magennis (forecasting with flow data), Jim Benson (Personal Kanban, Lean Coffee), Nigel Thurlow and John Turner (The Flow System), Chris Matts (real options), Michael Feathers (legacy code), Christopher Alexander (pattern languages)

Policy lens: Digital service teams that measure flow and recovery rather than output, fund products not projects, and design public platforms to keep working when conditions change.

Engage: [DORA research program](#) · [The Flow System](#) · [DevOps Research and Assessment \(overview\)](#)

Go deeper: [The Complexity Lounge](#) · [DORA \(DevOps Research and Assessment\)](#)

Cases: A decade of research on software delivery (see Cases)

Learning and pedagogy

Experiential learning and reflective practice Method

Learning as a cycle of doing, noticing, making sense, and trying again. The base layer for learning complexity, because complexity is mostly learned by acting and reflecting.

Road: John Dewey → Kurt Lewin → David Kolb (the experiential learning cycle) → Donald Schön (reflection-in-action)

Mix in: Action research, T-groups, improvement cycles

Policy lens: Learning built into delivery: reflective supervision for frontline staff, learning reviews after each cycle, and training that happens on real cases.

Engage: [David A. Kolb on experiential learning \(infed\)](#) · [Kurt Lewin: groups, experiential learning, and action research \(infed\)](#)

Go deeper: [NTL Institute](#) · [infed \(YMCA George Williams College\)](#)

Cases: Outward Bound · Phenomenon-based learning in Finland's core curriculum (see Cases)

Transformative learning Lineage

Learning that changes the frame, not just the content. Often starts with a disorienting dilemma and moves through critical reflection toward a new perspective. Threshold concepts name the ideas that are troublesome before they become obvious.

Road: Jack Mezirow (perspective transformation) → Edward Taylor and Patricia Cranton → Jan Meyer and Ray Land (threshold concepts and troublesome knowledge)

Mix in: Freire, Robert Kegan, Nora Bateson on education

Policy lens: Designing public-sector learning so staff can revisit core assumptions, not just learn new procedures, especially when a service model is changing.

Engage: [Taylor: The Theory and Practice of Transformative Learning \(ERIC\)](#) · [The Value of Troublesome Knowledge \(Composition Forum\)](#)

Go deeper: [International Transformative Learning Association](#)

Cases: Women's re-entry programs study (see Cases)

Critical and popular education Hybrid

Education as practice of freedom: learners read their own world, name the forces shaping it, and act together. Popular education schools trained the movements that changed laws.

Road: Paulo Freire (*Pedagogy of the Oppressed*, conscientization) → Myles Horton and the Highlander Folk School → bell hooks (*Teaching to Transgress*, engaged pedagogy)

Mix in: Orlando Fals-Borda, Grace Lee Boggs, power analysis

Policy lens: Community education that builds residents' own analysis before consultation, and civic learning that treats people as co-authors of policy.

Engage: [Paulo Freire: dialogue, praxis and education \(infed\)](#) · [bell hooks on education \(infed\)](#) · [How Myles Horton built a social movement institution \(Convergence\)](#)

Go deeper: [James and Grace Lee Boggs Center](#) · [infed \(YMCA George Williams College\)](#) · [Highlander Research and Education Center](#)

Cases: Highlander Folk School and the civil rights movement · People's Plan Campaign · Participatory action research on Colombia's Caribbean coast (see Cases)

Communities of practice and action learning Structure

Learning held by groups of practitioners over time. Communities of practice share a domain; action learning sets meet to work on real problems through questions rather than advice.

Road: Reg Revans (action learning, learning equals programmed knowledge plus questioning) → Jean Lave and Etienne Wenger (communities of practice) → Etienne and Beverly Wenger-Trayner (social learning, systems convening)

Mix in: Art of Hosting, Human Learning Systems, developmental evaluation

Policy lens: Cross-agency practitioner communities, peer learning networks between cities, and action learning sets for managers facing problems with no known answer.

Engage: [Wenger-Trayner: Introduction to communities of practice](#) · [Revans Action Learning archive \(University of Salford\)](#)

Go deeper: [Wenger-Trayner](#)

Cases: Xerox Eureka (see Cases)

Adult development and immunity to change Lineage

How adults grow in their capacity to hold complexity, and why people stay stuck even when they want to change. Stages describe how someone makes meaning, not how much they are worth.

Road: Jean Piaget → Robert Kegan (*In Over Our Heads*) → Kegan and Lisa Lahey (immunity to change) → Jennifer Garvey Berger (*Simple Habits for Complex Times*)

Mix in: Transformative learning, Group Relations, Theory U

Policy lens: Leadership programs that work with competing commitments behind stalled reforms, without turning developmental stages into a ranking of staff.

Engage: [Kegan and Lahey: The Real Reason People Won't Change \(HBR\)](#) · [Simple Habits for Complex Times \(Cultivating Leadership\)](#)

Go deeper: [Cultivating Leadership](#)

Cases: Deliberately developmental organizations (see Cases)

Complexity and education Lineage

Treating classrooms, schools, and learning programs as complex systems in their own right, and asking what it takes to educate people for a world that cannot be predicted.

Road: Edgar Morin (*Seven Complex Lessons in Education for the Future*) → Brent Davis and Dennis Sumara (*Complexity and Education*) → Nora Bateson on education → Inner Development Goals (a widely used capacity framework)

Mix in: The Systems Change Learning Guide's learning areas, communities of practice, threshold concepts

Policy lens: Public education and workforce programs that teach uncertainty, relationship, and systems literacy alongside content, and that evaluate learning developmentally.

Engage: [Morin: Seven complex lessons in education for the future \(UNESCO\)](#) · [Nora Bateson: Preparing for a confusing future](#) · [Inner Development Goals guide](#)

Go deeper: [International Bateson Institute](#) · [Inner Development Goals](#) · [Systems Change Learning Guide](#)

Cases: Phenomenon-based learning in Finland's core curriculum (see Cases)

Cases

Real situations where a pathway met the world. Each names its limits or later reversals, because the learning is in both.

Project Cybersyn Chile · 1971 to 1973

Stafford Beer's team built a real-time economic feedback network for Salvador Allende's government using the Viable System Model, telex machines, and an operations room.

Lesson: Cybernetic governance can move information and authority closer to where work happens. It also shows how quickly politics overtakes design.

Limits: Ended by the 1973 coup. Historians still debate how much it worked as intended.

Pathways: Cybernetics, the shared foundation, Governance and self-organizing structures

Sources: [Eden Medina: Cybernetic Revolutionaries \(MIT Press\)](#)

Risk Assessment and Horizon Scanning Singapore · 2004 onward

Singapore's government built a national horizon-scanning program informed by Cynefin and narrative sensemaking to detect weak signals early.

Lesson: Sensemaking methods can be institutionalized inside a state, but they need continual protection from reverting to prediction.

Limits: Later commentary asks how much of the original ambition survived.

Pathways: Sensemaking and acting in uncertainty, Strategy under uncertainty

Sources: [The Cynefin Co: Horizon scanning](#) · [Integrating foresight into policy: RAHS in Singapore](#)

Buurtzorg neighbourhood nursing Netherlands · 2006 onward

Home care delivered by small self-managing nurse teams with minimal overhead and no middle management, scaling to thousands of nurses.

Lesson: Distributed authority can scale when support functions serve teams instead of managing them.

Limits: Results depend on Dutch funding rules and professional nursing culture; transplants elsewhere have been mixed.

Pathways: Governance and self-organizing structures, Corporate innovation and renewal, Teams and learning cultures

Sources: [Commonwealth Fund case study](#) · [Centre for Public Impact profile](#)

Amsterdam City Doughnut Amsterdam, Netherlands · 2020 onward

The city adopted a Doughnut portrait to guide its circular economy strategy, asking how the city can thrive within planetary and social boundaries.

Lesson: A shared picture of the goal can realign many departments without a single master plan.

Limits: Researchers question how far it has changed core economic decisions.

Pathways: Complexity economics and new economic models, Public policy and government innovation

Sources: [DEAL: Amsterdam City Doughnut](#) · [TIME: Amsterdam is embracing a new economic theory](#)

Participatory budgeting Porto Alegre, Brazil · 1989 onward

Residents deliberated and voted on a share of the city's capital budget through neighborhood and thematic assemblies, and the model spread worldwide.

Lesson: Real decision power, not consultation, draws people in. Participation shifted spending toward poorer neighborhoods.

Limits: Weakened by later administrations; recent research documents its dismantling in its birthplace.

Pathways: Hosting and participatory practice, Participatory design and design justice, Power analysis

Sources: [WRI: Porto Alegre and the challenge of sustaining transformative change](#) · [Participedia: Porto Alegre 1989 to present](#) · [From innovation to erosion \(2026\)](#)

Child nutrition through positive deviance Vietnam · 1990 onward

Jerry and Monique Sternin worked with villages to find poor families whose children were well nourished, then helped others adopt their feeding practices.

Lesson: Solutions already present in a community spread faster than imported ones, because they come from peers.

Pathways: Positive deviance

Sources: [The Vietnam story: 25 years later](#)

100,000 Lives Campaign United States · 2004 to 2006

IHI enrolled thousands of hospitals in six evidence-based changes, using improvement cycles and peer networks to spread them.

Lesson: A campaign structure plus local PDSA testing can spread practice fast across independent organizations.

Limits: Its headline lives-saved figure was challenged by researchers, and IHI published a reply.

Pathways: Improvement and adaptive management

Sources: [IHI: 100,000 Lives Campaign ten years later](#) · [The 100,000 Lives Campaign: a scientific and policy review \(PubMed\)](#)

Menominee Forest Wisconsin, United States · 1854 onward

The Menominee Nation has managed its forest on a sustained-yield basis for generations, harvesting while the standing timber has grown.

Lesson: Long-horizon stewardship rooted in relationship with land can outperform short-cycle management on both ecology and livelihood.

Pathways: Indigenous and land-based knowledge, Living systems and regeneration

Sources: [Yale Environment 360: Menominee forest management](#) · [College of Menominee Nation: an Indigenous model of sustainable development \(Sustainability Science, open access\)](#)

Student Volunteer Army Christchurch, Aotearoa New Zealand · 2010 onward

After the Canterbury earthquakes, students self-organized thousands of volunteers to clear liquefaction silt, then became a lasting emergency response organization.

Lesson: Spontaneous volunteers become an asset when officials make room for them and when light structure forms quickly.

Pathways: Crisis, disaster, and high reliability, Emergence in movements and social change

Sources: [A shovel or a shopping cart: ten years of the SVA \(PMC\)](#)

Plymouth Alliance for Complex Needs Plymouth, England · 2019 onward

The council and health commissioners replaced competing service contracts with a single alliance of providers sharing budget and responsibility for people with complex needs.

Lesson: Contract form shapes behavior. Alliance commissioning lets providers learn together instead of competing over targets.

Pathways: Government plumbing, Public policy and government innovation, Funding and budgeting for adaptation

Sources: [HLS case study: Plymouth Alliance](#) · [Co-operative Councils Innovation Network case study](#)

Human Learning Systems in practice UK and beyond · 2019 onward

Dozens of public service organizations documenting their shift from targets and payment by results toward learning, relationships, and system stewardship.

Lesson: The shift is less about a new tool than about changing funding, accountability, and management at the same time.

Pathways: Public policy and government innovation, Measurement and accountability, Developmental evaluation

Sources: [HLS case studies](#) · [JRF: relational public service can tackle hardship](#)

Legible Practises Helsinki, Finland · 2013

A free book of six stories, collected by Helsinki Design Lab, about public institutions that changed how they work and the stewardship craft behind it.

Lesson: Stewardship is a craft that can be described and taught when practitioners make their decisions visible.

Pathways: Strategic design and policy labs, Public policy and government innovation

Sources: [Legible Practises \(free book\)](#)

Water shortage in Ojai Ojai, California · 2016 onward

Carnegie Mellon's transition design team worked with residents and stakeholders to map a wicked water problem and seed interventions across timescales.

Lesson: Mapping a problem with the people inside it, across long horizons, reveals where small interventions can connect.

Pathways: Transition design

Sources: [Words into action: transition design in Ojai \(SciELO\)](#)

Handelsbanken and Equinor without annual budgets Sweden and

Norway · 1970s onward

Two large firms dropped fixed annual budgets for rolling forecasts, relative targets, and local authority.

Lesson: Budget processes are a governance structure; changing them changes what people are allowed to learn.

Pathways: Funding and budgeting for adaptation, Corporate innovation and renewal

Sources: [Forbes: how Equinor and Handelsbanken abolished budgeting](#) · [Handelsbanken case story \(Agile Business Consortium\)](#)

Highlander Folk School and the civil rights movement Tennessee,

United States • 1932 onward

Workshops where ordinary people analyzed their own situations trained labor organizers and civil rights leaders, including Rosa Parks and Septima Clark's citizenship schools.

Lesson: Popular education builds the analysis and relationships that movements later act on.

Pathways: Critical and popular education, Emergence in movements and social change

Sources: [How Myles Horton built a social movement institution](#) · [Highlander Folk School \(King Institute, Stanford\)](#)

Haier's RenDanHeYi China • 2005 onward

The appliance maker broke itself into thousands of microenterprises that contract with each other and serve users directly.

Lesson: Large organizations can shrink the distance between decisions and users by dissolving middle layers.

Pathways: Corporate innovation and renewal

Sources: [Corporate Rebels: RenDanHeYi](#)

Wajir Peace and Development Committee Wajir, Kenya • 1992 onward

After clan violence killed hundreds, women traders led by Dekha Ibrahim Abdi began mediating between clans. The agreement grew into a standing committee of elders, religious leaders, security services, officials, and NGOs.

Lesson: Peace held when people from every side kept meeting to carry it out, not when a single agreement was signed. Traditional Somali ways of settling disputes did much of the work.

Limits: The model spread to other districts in northern Kenya; violence in the region has flared again since.

Pathways: Conflict transformation, Emergence in movements and social change

Sources: [Right Livelihood: Dekha Ibrahim Abdi \(2007 laureate\)](#) · [Dekha Ibrahim Abdi \(Wikipedia\)](#)

Truth and Reconciliation Commission South Africa • 1996 to 2003

Public hearings across the country let victims of apartheid-era violence tell their stories and let perpetrators apply for amnesty in exchange for full disclosure.

Lesson: A shared public ritual can mark a passage between two orders of society and help people move through it together.

Limits: Many survivors felt reparations and prosecutions never followed through; the TRC remains debated as a model.

Pathways: Ritual, liminality, and transition, Conflict transformation

Sources: [Truth and Reconciliation Commission \(Department of Justice, South Africa\)](#) · [Truth and Reconciliation Commission \(Wikipedia\)](#)

Mont Fleur scenarios South Africa • 1991 to 1992

As negotiations to end apartheid began, 22 leaders from the ANC, the government, business, unions, and academia built four stories of where the country could go, named Ostrich, Lame Duck, Icarus, and Flight of the Flamingos.

Lesson: Scenarios built by people who disagree can give a whole country a shared language for choices ahead, without forcing agreement on what to do.

Limits: Participants and facilitator Adam Kahane describe its influence; it is hard to separate from everything else happening in the transition.

Pathways: Systemic design and futures, Strategy under uncertainty

Sources: [Reos Partners: Lessons from the Mont Fleur scenarios \(PDF\)](#) · [The Mont Fleur scenarios \(Open Foresight Hub\)](#)

Sustainable Food Lab US, Europe, and Latin America • 2004 onward

About 30 leaders from food companies, NGOs, and governments went on a two-year learning journey designed around Theory U, with field visits, a wilderness retreat, and prototype projects.

Lesson: People who compete or oppose each other can sense a system together first and then act on what they saw, but prototypes need a home after the lab ends.

Limits: Became a lasting network; some prototypes stalled and the founders write openly about trial and error.

Pathways: Presencing, Theory U, and Social Presencing Theater, Collaboration across boundaries

Sources: [Reos Partners: The Sustainable Food Lab, eight years and counting](#) · [Hal Hamilton: Trial and Error, a personal story \(PDF\)](#)

People's Plan Campaign Kerala, India • 1996 to 2001

The state devolved 35 to 40 percent of its development budget to local governments. Village assemblies in every ward set priorities, and volunteers trained in large numbers helped draw up local plans.

Lesson: Moving money and authority closer to people works when it comes with mass learning, so residents can actually use the power they are given.

Limits: Quality varied by place; later governments kept the structures but some of the campaign energy faded.

Pathways: Public policy and government innovation, Critical and popular education

Sources: [Isaac and Franke: Local Democracy and Development \(book summary\)](#) · [Building local democracy: evaluating decentralization in Kerala \(World Development\)](#)

Imagine Chicago Chicago, United States • 1992 onward

Young people, paired with adult mentors, interviewed business, civic, and cultural leaders about what gives the city life and what they hoped for, using Appreciative Inquiry questions.

Lesson: Asking what already works, and letting young people ask, can open conversations that problem-focused planning shuts down.

Limits: Results are mostly told as stories by the organizers; the model spread to other cities.

Pathways: Appreciative Inquiry

Sources: [Imagine Chicago: the early story](#) · [Imagine Chicago \(Appreciative Inquiry Commons\)](#)

Government Digital Service and GOV.UK United Kingdom · 2011 onward

A new team in the Cabinet Office replaced hundreds of government websites with one, GOV.UK, built around user needs, user research, and small multidisciplinary teams working in the open.

Lesson: Service design changes government when it comes with the authority to say no to old contracts and the skills to build in-house.

Limits: Its influence peaked in the early 2010s; later reviews describe a loss of central authority and uneven adoption.

Pathways: Service design, Government plumbing

Sources: [UK Government Digital Service: moving beyond a website \(Harvard Kennedy School case\)](#) · [The UK government's digital transformation: how did it come about? \(Economics Observatory\)](#)

Participatory action research on Colombia's Caribbean coast

Córdoba, Colombia · 1972 to 1975

Orlando Fals Borda and La Rosca worked with the national peasant users' association to research land struggles with peasants, not about them, and returned the findings in forms people could use.

Lesson: Research can be a tool for the people it studies when they set the questions and own the results.

Limits: Historians, notably Joanne Rappaport, show how messy and contested the early practice was.

Pathways: Action research, Critical and popular education

Sources: [The participatory research of Orlando Fals Borda \(Participatory Methods, IDS\)](#) · [Joanne Rappaport: Cowards Don't Make History \(Duke University Press\)](#)

Deliberately developmental organizations United States · Studied 2011 to 2016

Robert Kegan and Lisa Lahey studied companies such as Decurion that build personal growth into daily work, for example by making weaknesses and mistakes openly discussable.

Lesson: Adult development can be part of how work is organized, not just a training course.

Limits: The evidence comes from a small number of firms studied by the theory's authors.

Pathways: Adult development and immunity to change

Sources: [Kegan, Lahey et al.: Making Business Personal \(HBR, April 2014\)](#) · [On deliberately developmental organizations \(Harvard Graduate School of Education\)](#)

Embrace infant warmer Stanford, United States, and India · 2007 onward

Students in Stanford's Design for Extreme Affordability class were asked for a cheaper incubator. Field visits showed many babies died far from hospitals, so they designed a warmer that works without steady power.

Lesson: Watching how people actually live can change the question you are answering, not just the answer.

Limits: Embrace later had to change its business model to reach scale; design alone did not solve distribution.

Pathways: Design thinking

Sources: [Embrace Innovations \(Wikipedia\)](#)

Strive Partnership Cincinnati and Northern Kentucky, United States · 2006 onward

School districts, universities, funders, businesses, and nonprofits agreed on shared goals and measures for children from cradle to career, with a backbone team to coordinate them.

Lesson: Organizations that each own one piece of a problem can coordinate when they share data and a common agenda.

Limits: The case that launched the Collective Impact model; later critiques say it underplayed community voice and power.

Pathways: Collaboration across boundaries, Measurement and accountability

Sources: [Kania and Kramer: Collective Impact \(SSIR, 2011\)](#) · [StriveTogether: reinventing the local education ecosystem \(Harvard Business School case, PDF\)](#)

Reframing Aging United States · 2015 onward

FrameWorks researched how Americans think about getting older, then worked with aging organizations to change the words and metaphors they used, such as 'older people' and 'building momentum'.

Lesson: Narrative change is slow, research-led, and collective: many organizations telling a new story the same way.

Limits: FrameWorks links the shift to later policy, such as Colorado's 2023 Job Application Fairness Act; attribution is its own.

Pathways: Narrative change

Sources: [FrameWorks: Impact spotlight, aging and ageism](#) · [Finding the Frame: research report \(FrameWorks, 2017, PDF\)](#)

Artificial Anasazi Long House Valley, Arizona, United States · Model published 2002

Researchers at the Santa Fe Institute and Brookings built an agent-based model of Ancestral Puebloan households from 800 to 1350 and compared it with archaeological records.

Lesson: Simple rules for many varied agents can reproduce real population patterns, and show which explanations for a collapse hold up.

Limits: The model matched growth well but not the final abandonment, suggesting social reasons beyond the environment.

Pathways: The formal science of complex systems

Sources: [Axtell et al.: Population growth and collapse in a multiagent model \(PubMed\)](#) · [Artificial Anasazi model code \(CoMSES\)](#)

Outward Bound Aberdyfi, Wales · 1941 onward

Shipping owner Lawrence Holt and educator Kurt Hahn started month-long courses of sailing, expeditions, and rescue training for young seamen during the Second World War.

Lesson: Learning by doing hard things together, then reflecting on them, builds capacities a classroom can't.

Pathways: Experiential learning and reflective practice

Sources: [Outward Bound \(Wikipedia\)](#) · [Outward Bound USA: history](#)

Women's re-entry programs study New York, New Jersey, California, and Washington, United States · 1975 to 1978

Jack Mezirow and Victoria Marsick studied 83 women returning to college in 12 programs and found many went through a 'perspective transformation' in how they saw themselves and their roles.

Lesson: Adults change most deeply when a disorienting experience meets reflection and support from others going through the same thing.

Limits: The founding study of transformative learning; later scholars questioned how universal its ten phases are.

Pathways: Transformative learning

Sources: [Mezirow and Marsick: Education for Perspective Transformation \(Semantic Scholar\)](#) · [Jack Mezirow \(Wikipedia\)](#)

Xerox Eureka France and worldwide · 1990s

Anthropologist Julian Orr found copier technicians learned mostly by swapping stories. Xerox PARC then built Eureka, a peer-reviewed database of technicians' own repair tips, credited to their authors.

Lesson: Communities of practice already exist; support works best when it amplifies how people learn from each other.

Limits: Widely cited in knowledge management; most accounts come from Xerox and PARC.

Pathways: Communities of practice and action learning

Sources: [Eureka! Xerox discovers way to grow community knowledge \(KMWorld\)](#) · [Field work in the tribal office \(MIT Technology Review, 1998\)](#)

The New Britain workshop New Britain, Connecticut, United States · 1946

Kurt Lewin's team ran a workshop for community leaders on fair employment. When participants sat in on staff discussions of their own behavior, the group's here-and-now interaction became the learning.

Lesson: A group studying itself as it happens can learn faster than one studying a topic.

Limits: Led to the National Training Laboratories at Bethel, Maine, in 1947.

Pathways: T-groups and sensitivity training

Sources: [Ed Batista: A brief history of T-groups](#) · [Kurt Lewin: groups, experiential learning and action research \(infed\)](#)

The Leicester Conference Leicester, England · 1957 onward

The Tavistock Institute's two-week residential conference turns the event itself into a temporary organization, so members can study authority, leadership, and group life as they happen.

Lesson: Hidden group dynamics such as anxiety and projection become workable once they can be seen in the room.

Limits: Intense by design; some participants find it destabilizing.

Pathways: Group Relations

Sources: [Tavistock Institute: What is a group relations conference?](#) · [Tavistock Institute: The Leicester Conference](#)

Warning people 10,000 years from now United States · 1981 to 1993

The US Department of Energy's Human Interference Task Force asked how to mark nuclear waste sites so people thousands of years in the future would not dig there. Semiotician Thomas Sebeok's 1984 report, *Communication Measures to Bridge Ten Millennia*, treated it as a problem of signs that outlive their language.

Lesson: No single sign survives that long. Sebeok recommended mixing icons, indexes and symbols, building in heavy redundancy, and relaying the message, re-encoded by each generation, rather than trusting one monument.

Limits: The task force's recommendations were never implemented as such. Sandia's 1993 marker study for the Waste Isolation Pilot Plant built on this work; the final marker design is still being developed.

Pathways: Semiotics: signs, codes, and meaning-making, Strategy under uncertainty

Sources: [Science History Institute: Speaking to the future](#) · [Long Now: Communication measures to bridge 10 millennia](#)

European mink on Hiiumaa and Saaremaa Estonia · 2019 to 2021

Semioticians at the University of Tartu studied how island communities understood the European mink, an endangered species released on Hiiumaa since 2000, to help plan a further reintroduction on Saaremaa.

Lesson: An animal carries different meanings for different people. Reading those local meanings, alongside the species' own needs, reduces conflict in reintroduction.

Limits: A research and advisory project; the report gives recommendations rather than measured conservation outcomes.

Pathways: Biosemiotics and the Umwelt, Living systems and regeneration

Sources: [University of Tartu: Reintroduction of the European mink, analysis of the local community's experience](#) · [University of Tartu ecosemiotics group](#)

Dialogue at a steel mill United States · Early 1990s

The MIT Dialogue Project, led by William Isaacs, worked with union and management at a steel mill whose relationship had been adversarial for about fifty years. They met in dialogue, with no deal on the table, to look at how they thought about each other.

Lesson: Participants said they had learned to question the labels they applied to each other, and found they had more in common than they expected. Dialogue changed the relationship that later negotiations ran through.

Limits: Reported by the project's own lead, without independent evaluation or long-term follow-up. The source does not name the plant.

Pathways: Bohm Dialogue, Conflict transformation

Sources: [William Isaacs: Dialogue, the power of collective thinking \(The Systems Thinker\)](#) · [The art of dialogue \(MIT\)](#)

The Trojan Warrior Project United States · 1985

A six-month US Army Special Forces experiment that taught a group of Green Berets aikido, meditation and body awareness alongside their training. Richard Strozzi-Heckler was one of the civilian teachers.

Lesson: Body-based awareness can be taught as a skill inside a demanding institution, not just practised privately. It also shows how such practices get absorbed into whatever purpose the institution already has.

Limits: Results were mixed by the soldiers' own accounts, and no formal evaluation was published. Strozzi-Heckler's account is a first-person book.

Pathways: Somatic and embodied practice

Sources: [Richard Strozzi-Heckler \(Wikipedia\)](#) · [In Search of the Warrior Spirit \(North Atlantic Books\)](#)

Despair and empowerment work after Chernobyl Russia and Ukraine · 1990s

Joanna Macy took her workshops on despair and empowerment in the nuclear age to the former Soviet Union after the 1986 Chernobyl accident, including a 1992 workshop in Russia for people living with its effects.

Lesson: Naming grief and fear together, in communities living with long-term contamination, helped turn paralysis into the energy to act and care for each other.

Limits: Documented mainly in tributes and practitioner accounts; there is no independent study of outcomes.

Pathways: Work That Reconnects

Sources: [Tricycle: Remembering Joanna Macy](#) · [Believe.Earth: Joanna Macy, pain as inspiration](#)

Zero Step Warm Data Project United Nations (online) · 2021

Alongside the formal working groups of the UN High-Level Dialogue on Energy, UNDP and the International Bateson Institute co-chaired a Warm Data process, People Need People Online, to help participants see how energy access is tangled up with family, health, economy and culture.

Lesson: A Warm Data process can sit beside a formal policy process without replacing it, giving people a different way to perceive the same issue.

Limits: The public record is an event notice; we found no published account of what changed as a result.

Pathways: Nora Bateson and Warm Data

Sources: [UN: Zero Step Warm Data Project \(side event notice\)](#)

Te Awa Tupua, the Whanganui River Aotearoa New Zealand · 2017 onward

A Treaty of Waitangi settlement with Whanganui iwi recognised the river, from the mountains to the sea with its physical and metaphysical elements, as an indivisible living whole and a legal person. Two guardians, one named by iwi and one by the Crown, act as its human face.

Lesson: Law can be rewritten to start from relationship: the river is treated as an ancestor and a whole, not a set of resources divided among owners.

Limits: Implementation is slow and contested, and legal personhood is a Western tool that only partly expresses the iwi's relationship with the river.

Pathways: Indigenous and land-based knowledge, Relational, transcontextual, and post-humanist knowing

Sources: [Te Awa Tupua \(Whanganui River Claims Settlement\) Act 2017](#) · [Whanganui District Council: Te Awa Tupua](#)

Phenomenon-based learning in Finland's core curriculum Finland ·

2016 onward

Finland's national curriculum, in force from 2016, requires every school to run at least one multidisciplinary learning module each year, where pupils help plan and study a real theme across subject boundaries, alongside transversal competences.

Lesson: An education system can build in time to study whole, real-world problems without abandoning subjects.

Limits: Early media reports that Finland had scrapped subjects were wrong; how deeply schools use the modules varies.

Pathways: Complexity and education, Experiential learning and reflective practice

Sources: [Finnish National Agency for Education: New national core curriculum for basic education \(PDF\)](#) · [Phenomenon-based learning in Finland: a critical overview \(Cogent Education, 2024\)](#)

Well-being of Future Generations Act Wales · 2015 onward

Wales made long-term thinking a legal duty: public bodies must pursue seven linked well-being goals using five ways of working (long term, prevention, integration, collaboration, involvement), overseen by a Future Generations Commissioner.

Lesson: Treating environmental, social and cultural crises as one pattern can be written into law, so that decisions are judged against the long term and against each other.

Limits: How far it changes real decisions is debated, and its duties are hard to enforce in court.

Pathways: The metacrisis and meaning, Public policy and government innovation

Sources: [Well-being of Future Generations \(Wales\) Act 2015 \(legislation.gov.uk\)](#) · [Well-being of Future Generations \(Wales\) Act 2015 \(Wikipedia\)](#)

Chronic illness care in an Australian general practice Inner-city Australia

• 1985 to 2007

Researchers traced how one mid-sized general practice improved its care for chronic illness over 23 years, using interviews with all staff, documents, observation and recorded practice meetings.

Lesson: Care improved continuously, but not through planned system change: improvements emerged from unrelated efforts such as hiring staff for other reasons, financial pressure and technology bought for other problems. The authors found complex responsive processes described this better than planned change.

Limits: A single critical case, studied in depth; it shows a pattern, not a rule.

Pathways: Complex responsive processes

Sources: [Booth, Zwar and Harris, BMC Family Practice \(2013\)](#)

The loss of the space shuttle Columbia United States • 2003

Columbia broke apart on re-entry in February 2003, killing seven astronauts. Foam striking the wing at launch was the physical cause, and foam strikes had happened on earlier flights without disaster.

Lesson: The Columbia Accident Investigation Board concluded that 'the NASA organizational culture had as much to do with the accident as the foam'. Known warning signs had become accepted as normal, a pattern resilience and safety science now teaches people to look for.

Limits: Investigations name causes after the fact; the hard part is seeing drift while it is happening.

Pathways: Resilience engineering and safety science, Crisis, disaster, and high reliability

Sources: [CAIB press release, 26 August 2003 \(archived\)](#) · [Congressional Research Service synopsis of the CAIB report](#)

Systems thinking in Portsmouth housing repairs Portsmouth, England •

Around 2010

Portsmouth City Council's housing service, which looked after about 17,000 homes, met its targets while tenants complained about repairs. Redesigning the work around what tenants actually needed, rather than activity targets, changed the service.

Lesson: Repair times fell from 24 days to 7, satisfaction rose from 60% to 99%, and the same staff completed far more jobs, according to an account citing a Wales Audit Office report. Targets had hidden the 'failure demand' created by the system itself.

Limits: Results come from practitioner accounts and an audit report; the approach depends on leaders willing to drop targets.

Pathways: Systems thinking traditions: hard, soft, and critical, Government plumbing

Sources: [APM: How Portsmouth Council used systems thinking \(2019\)](#) · [AMED: The Vanguard Method](#)

The Dutch childcare benefits scandal Netherlands · 2005 to 2021

The Dutch tax authority wrongly accused about 26,000 parents of fraud over childcare benefits. An algorithmic risk-classification system used nationality as a risk factor, and families were pushed into debt. The government resigned in January 2021.

Lesson: An automated system built for efficiency amplified suspicion against people who were already marginalized, and few could appeal. The Dutch Data Protection Authority found the processing unlawful and discriminatory.

Limits: Algorithms were one part of a wider failure of law, politics and administration.

Pathways: Technology, AI, and complexity, Measurement and accountability

Sources: [Amnesty International: Xenophobic machines \(October 2021\)](#) · [Dutch childcare benefits scandal \(overview\)](#)

An EU field guide for managing crises with constraints European Union · 2021

Dave Snowden and Alessandro Rancati wrote a field guide for decision makers for the European Commission's Joint Research Centre. It turns constraint thinking, which Cynefin takes from Alicia Juarrero, into practical steps for crises: set and manage boundaries, build informal structures, keep options open, distribute engagement and keep assessing the changing landscape.

Lesson: The most concrete public translation of constraint-based thinking into guidance for governments that we have found so far: act on the constraints that shape what can happen, rather than issuing detailed plans.

Limits: Not as strong as the other cases yet. It is guidance written for decision makers, not a documented outcome; how far it has changed decisions is not yet reported, and we are watching how it plays out.

Pathways: Mind, life, and constraints, Sensemaking and acting in uncertainty

Sources: [Snowden and Rancati: Managing complexity \(and chaos\) in times of crisis \(EU Publications Office, 2021\)](#) · [Complexity Lounge #21: Alicia Juarrero, Cause and Constraints](#)

A decade of research on software delivery Global · 2013 onward

The DORA research program, founded by Nicole Forsgren, Jez Humble and Gene Kim and now part of Google Cloud, has surveyed tens of thousands of technology professionals in annual State of DevOps reports.

Lesson: Fast, stable software delivery and a high-trust, generative culture go together: DORA reports that 'a high-trust, generative culture predicts software delivery and organizational performance in technology'.

Limits: Survey-based research; DORA itself warns against using its metrics to rank individual teams.

Pathways: Software, flow, and architecture, Teams and learning cultures

Sources: [DORA research program](#) · [DORA: generative organizational culture](#)

Organization profiles

Organizations carrying the work, each with public articles, talks, podcasts, or courses about the concepts. Media links were found by web search in September 2026.

International Bateson Institute

Founded by Nora Bateson to carry transcontextual research and Warm Data work. Hosts Warm Data Lab trainings and a global network of hosts.

Pathways: Nora Bateson and Warm Data, Relational, transcontextual, and post-humanist knowing, Complexity and education

Article: [Warm Data: information that is alive](#)

Essay: [Nora Bateson: Warm Data \(2017\)](#)

Essay: [Nora Bateson: Preparing for a confusing future](#)

The Emergence Network

Bayo Akomolafe's network and home for his courses and gatherings, including We Will Dance With Mountains.

Pathways: Relational, transcontextual, and post-humanist knowing, Ritual, liminality, and transition

Essay: [Bayo Akomolafe: The times are urgent, let's slow down](#)

Podcast: [For The Wild: Bayo Akomolafe on slowing down in urgent times](#)

Course: [We Will Dance With Mountains \(Bayo Akomolafe's offerings\)](#)

Gesturing Towards Decolonial Futures

Arts and research collective co-founded by Vanessa Machado de Oliveira, working on how modernity's harms shape even our efforts to change it.

Pathways: Relational, transcontextual, and post-humanist knowing, The metacrisis and meaning, Ritual, liminality, and transition

Book page: [Hospicing Modernity](#)

Podcast: [Green Dreamer: Vanessa Machado de Oliveira on sensing into collapse](#)

Center for Native Peoples and the Environment

Founded by Robin Wall Kimmerer at SUNY ESF to bring Indigenous and scientific knowledge together in service of shared goals.

Pathways: Indigenous and land-based knowledge

Essay: [Robin Wall Kimmerer: The Serviceberry \(Emergence Magazine\)](#)

Interview: [Robin Wall Kimmerer \(Yale Environment 360\)](#)

Interview: [Tyson Yunkaporta: Deep Time Diligence \(Emergence Magazine\)](#)

Presencing Institute

Otto Scharmer's action research platform for Theory U, u.lab, and Social Presencing Theater, used by public, civic, and corporate teams.

Pathways: Presencing, Theory U, and Social Presencing Theater

Article: [Theory U: leading from the emerging future](#)

Video: [An embodiment practice from Social Presencing Theater, with Arawana Hayashi](#)

Article: [Arawana Hayashi: Social Presencing Theater](#)

Perspectiva

London-based collective led by Jonathan Rowson working on the relationship between systems, souls, and society.

Pathways: The metacrisis and meaning

Essay: [Jonathan Rowson: Tasting the Pickle, ten flavours of meta-crisis](#)

Video: [RSA Animate: Iain McGilchrist, The Divided Brain](#)

The Cynefin Company

Dave Snowden's company for Cynefin, SenseMaker, and anthro-complexity, with trainings, retreats, and an open practitioner wiki.

Pathways: Sensemaking and acting in uncertainty

Article: [Snowden and Boone: A Leader's Framework for Decision Making \(HBR\)](#)

Article: [Horizon scanning \(from the Singapore RAHS work\)](#)

Human Systems Dynamics Institute

Founded by Glenda Eoyang. Teaches Adaptive Action and the CDE model through a global practitioner network.

Pathways: Sensemaking and acting in uncertainty

Article: [Introduction to Adaptive Action](#)

Video: [Glenda Eoyang explains human systems dynamics and adaptive action](#)

Sociocracy For All

Nonprofit teaching sociocracy (consent, circles, double links) to cooperatives, communities, and organizations, with free resources.

Pathways: Governance and self-organizing structures

Guide: [What is sociocracy?](#)

Video: [What is sociocracy? \(20 minute animation\)](#)

Ostrom Workshop

Indiana University research center founded by Elinor and Vincent Ostrom, continuing work on polycentric governance and the commons.

Pathways: Governance and self-organizing structures, Complexity economics and new economic models

Lecture: [Elinor Ostrom: Beyond Markets and States \(Nobel lecture\)](#)

American Society for Cybernetics

Home of second-order cybernetics in the US, with conferences that run as participatory conversations.

Pathways: Cybernetics, the shared foundation

Article: [Heinz von Foerster's ethical imperative](#)

Book page: [Eden Medina: Cybernetic Revolutionaries \(Project Cybersyn\)](#)

Santa Fe Institute

Independent research institute for complex systems science. Its Complexity Explorer offers free courses.

Pathways: The formal science of complex systems

Course: [Introduction to Complexity \(Complexity Explorer\)](#)

Stockholm Resilience Centre

Research center on social-ecological resilience, home of the planetary boundaries work.

Pathways: Living systems and regeneration

Article: [What is resilience?](#)

Capra Course

Fritjof Capra's online course on the systems view of life, with a live community of alumni.

Pathways: Living systems and regeneration

Course: [The Systems View of Life](#)

Essay: [Daniel Christian Wahl: Designing Regenerative Cultures, introduction](#)

Berkana Institute

Margaret Wheatley's institute behind the Two Loops model of how emerging systems grow as old ones decline.

Pathways: Emergence in movements and social change

Article: [Pioneering a new paradigm](#)

Article: [Chris Corrigan: The two loops model of change](#)

James and Grace Lee Boggs Center

Detroit center carrying the Boggses' visionary organizing: building community-based alternatives while resisting.

Pathways: Emergence in movements and social change, Critical and popular education

Essay: [The how and why of visionary organizing](#)

Interview: [adrienne maree brown: Earning our place on the planet \(Longreads\)](#)

Art of Hosting

Global community of practice for participatory leadership, carrying Open Space, World Café, circle, and chaordic design.

Pathways: Hosting and participatory practice

Article: [Chris Corrigan: What is the Art of Hosting?](#)

Article: [Chris Corrigan: Dear Dave, a letter on hosting, complexity and practice](#)

Liberating Structures

Henri Lipmanowicz and Keith McCandless's open set of 33 microstructures that include everyone in shaping the next step.

Pathways: Hosting and participatory practice

Article: [Keith McCandless: Liberating Structures, change methods for everybody every day](#)

Podcast: [Leadermorphosis: McCandless and Lipmanowicz on Liberating Structures](#)

NTL Institute

Founded in 1947 from Kurt Lewin's work; the origin of the T-group and much of organization development.

Pathways: T-groups and sensitivity training, Experiential learning and reflective practice

Article: [NTL: Kurt Lewin's legacy](#)

Article: [Kurt Lewin: groups, experiential learning and action research \(infed\)](#)

Tavistock Institute of Human Relations

London institute for applied social science; runs the Leicester Conference on authority, leadership, and organization.

Pathways: Group Relations

Article: [What is a Group Relations conference?](#)

Article: [The Leicester Conference](#)

Work That Reconnects Network

Carries Joanna Macy's practice after her death in 2025, with facilitators and gatherings worldwide.

Pathways: Work That Reconnects, Ritual, liminality, and transition

Article: [What is the Work That Reconnects?](#)

Article: [The Work That Reconnects \(Tricycle\)](#)

Fowler Center, Case Western Reserve University

Home of David Cooperrider's Appreciative Inquiry work and the AI Summit method for whole-system change.

Pathways: Appreciative Inquiry

Article: [Appreciative Inquiry at the Fowler Center](#)

Human Learning Systems

Network and body of practice led by Toby Lowe and colleagues, moving public services from targets toward learning, relationships, and system stewardship.

Pathways: Public policy and government innovation, Measurement and accountability, Government plumbing, Developmental evaluation

Report: [HLS reports and guides](#)

Article: [Managing complexity with Human Learning Systems \(i2Insights\)](#)

Case studies: [HLS case studies](#)

Centre for Public Impact

Works with governments on relational, learning-based public services and co-produced some of the key HLS reports.

Pathways: Public policy and government innovation

Report: [Human Learning Systems: a complexity-friendly approach to public services](#)

Case: [Buurtzorg: revolutionising home care in the Netherlands](#)

UCL Institute for Innovation and Public Purpose

Mariana Mazzucato's institute for mission-oriented policy and rethinking the state's role in shaping markets.

Pathways: Public policy and government innovation, Complexity economics and new economic models

Article: [Mariana Mazzucato: Policy with a Purpose \(IMF F&D, 2024\)](#)

Nesta

UK innovation foundation with long-running work on public innovation, collective intelligence, and anticipatory regulation.

Pathways: Public policy and government innovation, Government plumbing

Report: [Geoff Mulgan: Big Mind, how collective intelligence can change our world](#)

Report: [Renewing Regulation: anticipatory regulation in an age of disruption](#)

UNDP Accelerator Labs

A network of labs across UNDP country offices using sensemaking, experimentation, and portfolio approaches.

Pathways: Public policy and government innovation, Funding and budgeting for adaptation

Article: [Locally grown missions: UNDP's portfolio approach to system transformation](#)

Dark Matter Labs

Strategic design lab working on the institutional dark matter under cities: finance, contracts, governance, and ownership.

Pathways: Strategic design and policy labs, Public policy and government innovation

Essay: [Radicle Civics: building proofs of possibility for a civic economy](#)

Helsinki Design Lab (Sitra, archived)

Sitra's strategic design lab (2009 to 2013). Its free books remain a reference for government design studios.

Pathways: Strategic design and policy labs

Book: [Legible Practises: six stories about the craft of stewardship](#)

Article: [Helsinki Design Lab ten years later \(She Ji\)](#)

Doughnut Economics Action Lab

Kate Raworth's lab turning Doughnut Economics into practice with cities, businesses, and educators.

Pathways: Complexity economics and new economic models

Video: [Kate Raworth: A healthy economy should be designed to thrive, not grow \(TED\)](#)

Case: [Amsterdam City Doughnut](#)

International Futures Forum

Scottish collective behind the Three Horizons framework and practices for acting in powerless-feeling situations.

Pathways: Systemic design and futures, Strategy under uncertainty

Book page: [Bill Sharpe: Three Horizons, the patterning of hope](#)

UNESCO Futures Literacy

UNESCO's program on the capability to use the future, including Futures Literacy Laboratories.

Pathways: Systemic design and futures

Article: [Unlocking the power of futures: insights from Riel Miller \(SOIF\)](#)

Reos Partners

Adam Kahane's firm for transformative scenario planning and stretch collaboration among people who disagree.

Pathways: Strategy under uncertainty, Presencing, Theory U, and Social Presencing Theater

Excerpt: [Adam Kahane: Transformative Scenario Planning \(SSIR\)](#)

Design Justice Network

Network of designers and organizers centering people most affected by design decisions, around ten shared principles.

Pathways: Participatory design and design justice

Guide: [Design Justice Network principles](#)

Transition Design (Carnegie Mellon)

Terry Irwin, Gideon Kossoff, and colleagues' approach to designing for long-horizon societal transitions.

Pathways: Transition design

Article: [Irwin: Transition Design, a proposal for a new area of design practice](#)

Case: [Words into action: transition design in Ojai, California](#)

Good Services

Lou Downe's practice and publications on what makes a public or private service good.

Pathways: Service design

Guide: [Lou Downe: 15 principles of good service design](#)

Article: [Introducing the Good Services Scale](#)

The W. Edwards Deming Institute

Keeps Deming's System of Profound Knowledge in circulation through courses, podcasts, and free material.

Pathways: Improvement and adaptive management, Measurement and accountability

Article: [The Deming System of Profound Knowledge](#)

Institute for Healthcare Improvement

Founded by Don Berwick. Spread the Model for Improvement and PDSA cycles across health systems worldwide.

Pathways: Improvement and adaptive management

Guide: [The Model for Improvement](#)

Article: [100,000 Lives Campaign: ten years later](#)

Tamarack Institute

Canadian institute for community change, with deep work on developmental evaluation and Collective Impact 3.0.

Pathways: Developmental evaluation, Collaboration across boundaries

Guide: [Jamie Gamble: A Developmental Evaluation Primer \(McConnell Foundation\)](#)

Article: [Michael Quinn Patton on developmental evaluation principles \(AEA365\)](#)

Webinar: [Collective Impact 3.0](#)

Positive Deviance Collaborative

Carries Jerry and Monique Sternin's approach of finding and spreading solutions communities already have.

Pathways: Positive deviance

Case: [The Vietnam story: 25 years later](#)

Stanford d.school

Hasso Plattner Institute of Design at Stanford, which popularized the design thinking process now taught worldwide.

Pathways: Design thinking

Guide: [Design Thinking Bootleg](#)

Critique: [Natasha Iskander: Design thinking is fundamentally conservative \(HBR\)](#)

Action Research Network of the Americas

Network linking action researchers and participatory action researchers across North and South America.

Pathways: Action research

Article: [About ARNA](#)

Chapter: [ARNA: constructing a new network of North-South convergence](#)

International Transformative Learning Association

Carries Jack Mezirow's legacy through the International Transformative Learning Conference and its proceedings.

Pathways: Transformative learning

Proceedings: [Past conferences and proceedings](#)

Review: [Taylor: The Theory and Practice of Transformative Learning \(ERIC\)](#)

Institute of Development Studies: Powercube

John Gaventa and colleagues' open resource for analyzing spaces, places, and forms of power.

Pathways: Power analysis

Guide: [What is the Powercube?](#)

Project: [Powercube: understanding power for social change \(IDS\)](#)

JASS (Just Associates)

Feminist movement support organization whose power frameworks, from A New Weave of Power onward, are widely taught.

Pathways: Power analysis

Guide: [Just Power: engaging and resisting](#)

Guide: [A New Weave of Power, People and Politics \(VeneKlasen and Miller, JASS\)](#)

FrameWorks Institute

Research organization that studies how the public thinks about social issues and how reframing changes what policies seem possible.

Pathways: Narrative change

Article: [Five questions about framing](#)

Narrative Initiative

Supports narrative change practice across movements and funders.

Pathways: Narrative change

Report: [Toward New Gravity](#)

Article: [A living narrative definition to promote social change](#)

Leading Change Network

Global network teaching Marshall Ganz's organizing and public narrative practice.

Pathways: Narrative change

Guide: [Public narrative: story of self, us, and now](#)

Kroc Institute for International Peace Studies

Notre Dame institute where John Paul Lederach's peacebuilding work is rooted.

Pathways: Conflict transformation

Podcast: [John Paul Lederach: The Art of Peace \(On Being\)](#)

Book page: [The Moral Imagination](#)

Process Work Institute

Teaches Arnold and Amy Mindell's Process Work, Deep Democracy, and Worldwork for large-group conflict.

Pathways: Conflict transformation

Article: [Organizations and Worldwork](#)

Article: [Amy and Arnold Mindell: Deep Democracy](#)

Zehr Institute for Restorative Justice

Eastern Mennonite University institute carrying Howard Zehr's restorative justice work.

Pathways: Conflict transformation

Book page: [The Little Book of Restorative Justice](#)

Summary: [Summary of The Little Book of Restorative Justice \(Beyond Intractability\)](#)

Cultural Somatics Institute and generative somatics

Resmaa Menakem's cultural somatics, alongside the legacy archive of generative somatics (Staci K. Haines), which brought somatics into social justice organizing.

Pathways: Somatic and embodied practice

Podcast: [Resmaa Menakem: Notice the rage, notice the silence \(On Being\)](#)

Podcast: [Staci K. Haines: somatics for trauma healing and transformative justice \(Green Dreamer\)](#)

Archive: [generative somatics legacy](#)

Natural Hazards Center

University of Colorado Boulder center linking disaster research and practice; runs the CONVERGE network for social science in disasters.

Pathways: Crisis, disaster, and high reliability

Network: [CONVERGE](#)

Interview: [Social capital and community resilience: a conversation with Daniel Aldrich](#)

Review: [Rebecca Solnit: A Paradise Built in Hell \(SSIR\)](#)

William Bridges Associates

Carries William Bridges's transition model for the human side of organizational change.

Pathways: Ritual, liminality, and transition

Article: [What is transition?](#)

Article: [Francis Weller: The Geography of Sorrow \(The Sun\)](#)

Wardley Mapping community

Simon Wardley's openly licensed book and the community around mapping how value chains evolve.

Pathways: Strategy under uncertainty

Book: [Wardley Maps \(free online\)](#)

Article: [Pierre Wack: Scenarios, Uncharted Waters Ahead \(HBR\)](#)

Trust-Based Philanthropy Project

Peer-to-peer initiative helping funders shift power toward grantees through multi-year, flexible, relational funding.

Pathways: Funding and budgeting for adaptation

Guide: [Trust-based philanthropy practices](#)

Article: [Building a trust-based philanthropy to shift power back to communities \(SSIR\)](#)

Recoding America

Jennifer Pahlka's book and ongoing work on why government delivery fails and how policy and implementation can meet.

Pathways: Government plumbing

Podcast: [Jennifer Pahlka on reshaping government \(WBUR On Point\)](#)

Review: [System Error: review of Recoding America \(SSIR\)](#)

Team Topologies

Matthew Skelton and Manuel Pais's approach to organizing teams for fast flow, widely used in digital government.

Pathways: Teams and learning cultures

Guide: [Team Topologies key concepts](#)

Article: [Martin Fowler: Team Topologies](#)

Collective Impact Forum

Community of practice for cross-sector collaboration, now centered on equity and community power.

Pathways: Collaboration across boundaries

Article: [Collective Impact 3.0: an evolving framework for community change](#)

Corporate Rebels

Researchers who document progressive, self-managing workplaces, including Haier and Buurtzorg.

Pathways: Corporate innovation and renewal

Article: [RenDanHeYi: the organizational model defining the future](#)

Article: [Haier: entrepreneurship at scale \(Michele Zanini\)](#)

infed (YMCA George Williams College)

Free encyclopedia of informal education with clear, well-sourced essays on the major educators.

Pathways: Experiential learning and reflective practice, Critical and popular education

Article: [David A. Kolb on experiential learning](#)

Article: [Paulo Freire: dialogue, praxis and education](#)

Article: [bell hooks on education](#)

Highlander Research and Education Center

Founded in 1932 as the Highlander Folk School by Myles Horton; trained leaders of the labor and civil rights movements.

Pathways: Critical and popular education

Article: [How Myles Horton built a social movement institution \(Convergence\)](#)

Article: [Highlander Folk School \(SNCC Digital Gateway\)](#)

Wenger-Trayner

Etienne and Beverly Wenger-Trayner's practice on communities of practice, social learning, and systems convening.

Pathways: Communities of practice and action learning

Article: [Introduction to communities of practice](#)

Cultivating Leadership

Jennifer Garvey Berger's firm working with adult development and complexity in leadership.

Pathways: Adult development and immunity to change

Book page: [Simple Habits for Complex Times](#)

Article: [Kegan and Lahey: The Real Reason People Won't Change \(HBR\)](#)

Inner Development Goals

Nonprofit framework of inner capacities for sustainable development. Widely used; read it alongside critiques of ranking inner growth.

Pathways: Complexity and education

Guide: [IDG guide](#)

Resources: [IDG resources](#)

Systems Change Learning Guide

Jaya Ramchandani and Raisa Mirza's open capacity framework: 24 learning areas, a five-phase route, and the live library this app reads.

Pathways: Complexity and education

Guide: [The map, route, and library](#)

International Society for Biosemiotic Studies

Founded in 2005 by Copenhagen and Tartu researchers to study sign processes in living systems. Runs the annual Gatherings in Biosemiotics (held since 2001) and is linked to the Springer journal Biosemiotics.

Pathways: Biosemiotics and the Umwelt

Journal: [Biosemiotics \(Springer\)](#)

Article: [Biosemiotics \(overview\)](#)

University of Tartu Department of Semiotics

Home of the Tartu-Moscow school of Yuri Lotman and of Tartu biosemiotics and ecosemiotics. Its researchers apply Umwelt analysis to city green spaces and to species reintroduction.

Pathways: Biosemiotics and the Umwelt, Semiotics: signs, codes, and meaning-making

Article: [Ecosemiotics research group](#)

Journal: [Sign Systems Studies \(open access\)](#)

Article: [Reintroduction of the European mink: local community experience \(project page\)](#)

International Association for Semiotic Studies

The worldwide association for semiotics, founded in 1969. Holds world congresses and links national societies such as the Semiotic Society of America.

Pathways: Semiotics: signs, codes, and meaning-making

Article: [Peirce's theory of signs \(Stanford Encyclopedia of Philosophy\)](#)

Article: [Semiotic Society of America](#)

The Complexity Lounge

A practitioner meetup, started in New York, that has posted 65 recorded talks on complexity science, Cynefin, constraints, flow and organizational change, with guests from Dave Snowden to Karl Friston.

Pathways: Sensemaking and acting in uncertainty, Mind, life, and constraints, Software, flow, and architecture, Resilience engineering and safety science, Complex responsive processes, Systems thinking traditions: hard, soft, and critical

Video: [Complexity Lounge #12: The Latest from the Cynefin Centre \(Dave Snowden\)](#)

Video: [Complexity Lounge #21: Cause and Constraints \(Alicia Juarrero\)](#)

Video: [Complexity Lounge #62: The Nature of Everything \(Karl Friston\)](#)

Video: [Complexity Lounge #54: Complexity and Change \(Christopher Mowles\)](#)

Video: [Complexity Lounge #41: Systems Thinking & The Management of Complexity \(Michael C. Jackson\)](#)

Video: [Complexity Lounge #42: A Talk on Hidden Events \(Ron Westrum\)](#)

Video: [Complexity Lounge #15: Complexity and Machine Learning \(Abeba Birhane\)](#)

Complexity and Management (University of Hertfordshire)

Blog of the staff group on Hertfordshire's Doctor of Management, which teaches organizations as complex responsive processes of relating through reflexive research on practitioners' own work.

Pathways: Complex responsive processes

Article: [About the blog](#)

Video: [Complexity Lounge #54: Complexity and Change \(Christopher Mowles\)](#)

Video: [Complexity Lounge #63: The Wiggly World of Organizations \(Chris Rodgers\)](#)

DORA (DevOps Research and Assessment)

Research program on software delivery performance and culture, founded by Nicole Forsgren, Jez Humble and Gene Kim and now part of Google Cloud.

Pathways: Software, flow, and architecture, Resilience engineering and safety science

Research: [DORA research program](#)

Guide: [Generative organizational culture](#)

Distributed AI Research Institute (DAIR)

Independent, community-rooted AI research institute founded by Timnit Gebru, studying the harms of AI and alternatives to it.

Pathways: Technology, AI, and complexity

Paper: [Birhane and Cummins: Algorithmic Injustices, towards a relational ethics](#)

Video: [Complexity Lounge #15: Complexity and Machine Learning \(Abeba Birhane\)](#)

Journeys

From crisis response to city innovation quiet-heron

quiet-heron started in government crisis response, working with small, ritual-rich communities, and later moved into a city innovation role inside a reluctant bureaucracy.

1. Starting point: crisis response in small communities Community and place, Group and dialogue, Body and person

What they already know: communities self-organize in a crisis, ritual holds people through disruption, and regulated bodies make better decisions. These pathways name and deepen that knowledge.

Pathways: Crisis, disaster, and high reliability, Ritual, liminality, and transition, Somatic and embodied practice, Indigenous and land-based knowledge, Positive deviance

2. Arriving in city government Organization, Society and policy

Sort ordered from complex problems so procedure can stay procedural. Read resistance as a system defense rather than a personal failing. Learn where procurement, budgets, and audit actually push back.

Pathways: Sensemaking and acting in uncertainty, Group Relations, Power analysis, Government plumbing

3. Doing the work Organization, Society and policy, Group and dialogue

Host Warm Data Labs alongside the structured methods, not upstream of them, to loosen fixed perception. Use service design and developmental evaluation for accountable, learning-based change. Build a community of practice with the people inside who already work this way.

Pathways: Nora Bateson and Warm Data, Service design, Developmental evaluation, Measurement and accountability, Communities of practice and action learning

The braid. Crisis and ritual knowing carried into the institution: the experience of communities adapting fast is proof of what the bureaucracy could do if its constraints loosened.

From corporate innovation to civic partnerships amber-finch

amber-finch ran an innovation team in a large company and now brokers partnerships between business, city government, and community groups.

1. Starting point: corporate innovation Organization

Comfortable with experiments, portfolios, and customer research. Less practiced at working where no one owns the problem.

Pathways: Corporate innovation and renewal, Strategy under uncertainty, Teams and learning cultures, Design thinking

2. Entering civic work Community and place, Society and policy

Learn why design thinking alone can flatten communities into users, and how power shapes who gets to define the problem.

Pathways: Collaboration across boundaries, Power analysis, Participatory design and design justice, Critical and popular education

3. Doing the work Society and policy, Group and dialogue

Convene rather than run. Move money toward trust and portfolios, measure learning instead of outputs, and help partners tell a shared story of why change is needed now.

Pathways: Hosting and participatory practice, Funding and budgeting for adaptation, Measurement and accountability, Narrative change

The braid. Business speed braided with civic patience: portfolio thinking is an asset in public work once it is paired with shared power.

Living guides and practices by level of focus

Levels of focus, not rank. These are the scales where each person or practice mostly works. None sits above another, and many move across levels.

Body and person	Resmaa Menakem, Staci K. Haines, Robert Kegan, Jennifer Garvey Berger. Practices: somatic work, immunity to change, reflective practice
Knowing and perception	Iain McGilchrist, Evan Thompson, Alicia Juarrero, Terrence Deacon, Isabelle Stengers, Donna Haraway, Karen Barad, Jan Meyer and Ray Land
Group and dialogue	Arawana Hayashi, William Isaacs, Chris Corrigan. Practices: Bohm Dialogue, T-groups, Group Relations, Warm Data Labs, Art of Hosting, Liberating Structures, Work That Reconnects, action learning
Organization	Dave Snowden, Glenda Eoyang, Otto Scharmer, Peter Senge, Amy Edmondson, David Cooperrider, Bjarte Bogsnes, Matthew Skelton and Manuel Pais. Practices: sociocracy, Appreciative Inquiry, Adaptive Action, Beyond Budgeting
Community and place	Nora Bateson, Bayo Akomolafe, adrienne maree brown, Robin Wall Kimmerer, Tyson Yunkaporta, Rebecca Solnit, Daniel Aldrich, Dean Spade. Practices: positive deviance, participatory action research, popular education, mutual aid
Society and policy	Toby Lowe, Geoff Mulgan, Hilary Cottam, Mariana Mazzucato, Jennifer Pahlka, Charles Sabel, Indy Johar, Lou Downe, Michael Quinn Patton, John Gaventa. Practices: Human Learning Systems, developmental evaluation, service design, narrative change
Planetary and civilizational	Vanessa Machado de Oliveira, Jonathan Rowson, Carl Folke, Daniel Christian Wahl, Fritjof Capra, Kate Raworth, Arturo Escobar, Yaneer Bar-Yam

Suggested braids

Nora Bateson + Otto Scharmer. Relational sensing beside a structured change process. The friction shows what each assumes about intention.

Bayo Akomolafe + Dave Snowden. Post-humanist inquiry beside practical sensemaking. Both distrust tidy plans for different reasons.

Robin Wall Kimmerer + Elinor Ostrom. Reciprocity beside commons governance. Two routes to the same insight about shared resources.

Service design + developmental evaluation + transition design. Journey, learning loop, and long horizon for government innovation.

Power analysis + Hosting. Hosting without power analysis can launder existing hierarchies; power analysis without hosting can stay academic.

Somatic practice + Crisis response. Regulated bodies make better decisions in the first hours, and recover better afterward.

Live resources in the companion app come from the public feed of the Systems Change Learning Guide by Jaya Ramchandani and Raisa Mirza (welearnwegrow.github.io/capacities), shown unmodified under CC BY-NC-ND 4.0.